

Research article

Characterization of multiple trichloroethene, *cis*-dichloroethene and 1,1-dichloroethene degrading propanotrophic communities

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ABSTRACT

Aerobic cometabolism offers a viable strategy for the remediation of chlorinated solvent plumes at oxic sites where anaerobic approaches are limited. Here, propane-enriched mixed cultures (derived from agricultural soils and an impacted site sediment) which previously degraded 1,4-dioxane, were evaluated for their capacity to also degrade trichloroethene (TCE), *cis*-1,2-dichloroethene (cDCE), and 1,1-dichloroethene (1,1-DCE) over successive transfers. Sustained biodegradation of TCE and cDCE was observed across multiple enrichments and cultures enriched on one compound generally degraded the other. In contrast, 1,1-DCE biodegradation was restricted to a subset of cultures and removal times increased over transfers. Further, 1,1-DCE removal was absent at elevated concentrations, both trends consistent with inhibitory or toxic effects. Whole genome sequencing analyses revealed pronounced substrate-dependent selection of microbial communities, with cDCE-degrading cultures being dominated by *Mycobacterium* and *Mycolicibacterium*, whereas TCE-degrading cultures were dominated by *Rhodococcus*. *Rhodococcus* metagenome-assembled genomes (MAGs) in the TCE degrading cultures classified as *R. opacus* or *R. wratislaviensis*. 1,1-DCE degrading cultures were dominated by *Pseudonocardia*, although the associated MAGs contained a truncated propane monooxygenase alpha subunit, suggesting other enzymes were responsible for 1,1-DCE transformation. Functional gene analysis identified both group 5 (*prmABCD*) and putative group 6 propane monooxygenases (although their expression was not examined). Together, these results demonstrate that substrate-specific pressures govern propanotrophic community structure and function, and highlight distinct roles of key actinobacterial genera in chlorinated ethene cometabolism. These findings support the development of propane-based bioaugmentation strategies for the treatment of mixed chlorinated solvent contamination under aerobic conditions.

1. Introduction

The chlorinated solvents, tetrachloroethene (PCE), trichloroethene (TCE) and 1,1,1-trichloroethane (TCA) have been among the most widely used cleaning and degreasing solvents in the United States (Doherty, 2000). Indiscriminate disposal of these chemicals has caused significant soil and water contamination and the remediation of such sites has been difficult and costly (McCarty, 2010). One approach to remediate chlorinated solvent contaminated sites involves bioremediation. PCE was first reported to be biologically reduced to TCE under anaerobic conditions (Bouwer et al., 1981). Similarly, TCE was later also reported to be biologically reduced to form *cis*-1,2-dichloroethene (cDCE) and vinyl chloride (VC) (Parsons and Lage, 1985; Vogel and McCarty, 1985) and TCA can be transformed into 1,1-dichloroethene (1,1-DCE) (Vogel and McCarty, 1987). In many cases, these chemicals

occur with 1,4-dioxane, which was used as a stabilizer in chlorinated solvent formulations (Adamson et al. 2014, 2015; Anderson et al., 2012). Bioremediation efforts for the chlorinated ethenes have involved biostimulation or bioaugmentation with microbial cultures containing *Dehalococcoides mccartyi* (Steffan and Vainberg, 2013). However, bioremediation with *D. mccartyi* is unlikely to be used at large, oxic sites because of the cost associated with creating anaerobic conditions. Although anaerobic 1,4-dioxane biodegradation has previously been reported (Shen et al., 2008), this contaminant is typically recalcitrant under anaerobic conditions (Ramalingam and Cupples, 2020). Another option, co-metabolic bioremediation, was first identified as a remediation method forty years ago (Hopkins and McCarty, 1995; Nelson et al., 1988; Pfiffner et al., 1997; Wilson and Wilson, 1985) and has gained recent interest due to the potential to treat large, dilute, mixed-contaminant, aerobic plumes (Hatzinger, 2022). The growing

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Table 1

Characteristics of MAGs (bins) that contained all four subunits of the propane monoxygenase (*prm*) operon. Some MAGs contained two subunits that were not annotated as *prm* subunits (*prmA**CDB*) (correct length and orientation). *Mycobacterium* in red contained both operons. MAGs with the highest percentage of mapped reads in each sample from this group are in bold. In four cases, all for *Pseudonocardia* (1,1-DCE samples, marked ⁺), *prmA* was truncated.

Sample	TCE/DCE/ Replicate	Bin	Mapped Genes (%)	Order	Lowest Classification	<i>prm</i> ABCD	<i>prmA</i> *CD*B	Completeness	Contamination	Length (Mb)
Site 1	TCE A	2	1.21	<i>Actinomycetales</i>	<i>Pseudonocardia broussonetiae</i>	X		98.51	2.75	7,491,913
Site 1	TCE A	19	42.90	<i>Actinomycetales</i>	<i>Mycobacterium</i>		X	94.83	0.38	5,751,466
Site 1	TCE B	2	0.95	<i>Actinomycetales</i>	<i>Pseudonocardia broussonetiae</i>	X		97.09	2.22	7,442,245
Soil 2	TCE A	13	61.50	<i>Actinomycetales</i>	<i>Rhodococcus wratislaviensis</i>	X		99.75	2.10	8,519,066
Soil 2	TCE B	15	56.14	<i>Actinomycetales</i>	<i>Rhodococcus wratislaviensis</i>	X		96.54	2.10	7,850,513
Soil 3	TCE A	18	6.66	<i>Actinomycetales</i>	<i>Rhodococcus opacus</i>	X		99.66	2.94	7,643,432
Soil 3	TCE B	N/A	N/A	N/A	No MAGs with <i>prm</i> operon	N/A	N/A	N/A	N/A	N/A
Soil 4	TCE A	6	5.87	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X		95.78	1.73	6,093,708
Soil 4	TCE A	11	34.65	<i>Actinomycetales</i>	<i>Rhodococcus opacus</i>	X		98.62	3.88	7,582,893
Soil 4	TCE A	27	1.47	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X		99.55	7.53	8,620,668
Soil 4	TCE B	10	2.84	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X		91.36	2.35	8,095,027
Soil 4	TCE B	24	24.75	<i>Actinomycetales</i>	<i>Rhodococcus opacus</i>	X		98.90	3.83	8,110,727
Site 1	cDCE A	1	30.16	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X		100.0	4.16	8,920,031
Site 1	cDCE B	11	49.12	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X		100.0	3.70	8,869,529
Soil 2	cDCE A	18	1.29	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X		95.20	1.57	6,189,856
Soil 2	cDCE A	20	6.16	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X		94.58	3.01	6,385,544
Soil 2	cDCE A	29	1.11	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X		96.09	4.93	8,607,536
Soil 2	cDCE B	12	2.63	<i>Actinomycetales</i>	<i>Mycobacterium</i>		X	100.0	1.73	6,081,161
Soil 2	cDCE B	30	21.36	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X	X	100.0	1.82	7,003,254
Soil 3	cDCE A	4	2.52	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X		94.75	0.31	6,092,218
Soil 3	cDCE A	40	11.95	<i>Burkholderiales</i>	<i>Burkholderiaceae_B</i>		X	90.64	16.38	3,980,249
Soil 3	cDCE B	20	1.20	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X		99.72	2.61	7,300,948
Soil 3	cDCE B	45	28.52	<i>Actinomycetales</i>	<i>Mycobacterium fluoranthennivorans</i>		X	100.0	5.17	7,395,616
Soil 4	cDCE A	7	2.97	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X	X	99.06	14.09	7,565,700
Soil 4	cDCE B	2	0.32	<i>Actinomycetales</i>	<i>Mycobacterium</i>		X	96.98	0.54	5,733,864
Soil 4	cDCE B	16	0.45	<i>Burkholderiales</i>	<i>Methylibium</i>	X		90.21	19.28	4,757,184
Soil 4	cDCE B	36	0.47	<i>Actinomycetales</i>	<i>Mycobacterium</i>		X	98.79	1.36	6,662,032
Soil 4	cDCE B	37	20.05	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X	X	96.98	1.64	6,619,523
Soil 2	1,1-DCE A	11	3.94	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X		99.47	1.0	6,712,811
Soil 2	1,1-DCE A	15	37.91	<i>Actinomycetales</i>	<i>Pseudonocardia</i>	X ⁺		96.87	2.14	5,867,615
Soil 2	1,1-DCE B	6	1.30	<i>Actinomycetales</i>	<i>Mycobacterium</i>	X		98.77	1.85	7,047,595
Soil 2	1,1-DCE B	11	51.47	<i>Actinomycetales</i>	<i>Pseudonocardia</i>	X ⁺		97.34	1.77	5,842,628
Soil 3	1,1-DCE A	15	44.12	<i>Actinomycetales</i>	<i>Pseudonocardia</i>	X ⁺		97.04	1.77	5,872,741
Soil 3	1,1-DCE B	9	38.42	<i>Actinomycetales</i>	<i>Pseudonocardia</i>	X ⁺		97.58	2.30	5,967,428

importance of this approach is illustrated by fourteen field-scale aerobic co-metabolic bioremediation studies (Skinner et al., 2024).

Co-metabolic oxidation involves fortuitous oxidation of chemicals by monoxygenases or dioxygenases (Alexander, 1994; Horvath, 1972; Wackett, 1996). Such enzymes are induced by growth substrates, such as methane and propane, and because of their lower specificity, can oxidize other compounds beside the growth substrates (Alexander, 1994; Wackett, 1996). Oxygenases introduce one or two oxygen atoms from molecular oxygen to the molecule, leading to products which can then be further mineralized (Karigar and Rao, 2011). Although much attention has focused on the ability of methanotrophs to degrade the chlorinated solvents, these microbes are unable to degrade 1,4-dioxane (both in pure and mixed cultures) (Hatzinger et al., 2017). In contrast, propanotrophs offer much potential to degrade multiple contaminants, including the chlorinated ethenes and 1,4-dioxane (Skinner et al., 2024).

Propanotrophs are microorganisms that can use propane as a sole source of carbon and energy (Steffan et al., 1997; Wackett et al., 1989). These microorganisms are of particular interest because they have been associated with the biodegradation of numerous organic contaminants (Cappelletti et al., 2018; Hand et al., 2015; Sharp et al., 2007, 2010; Steffan et al., 1997; Wackett et al., 1989; Wang and Chu, 2017). For example, an early study documented TCE biodegradation by five *Mycobacterium* strains growing on propane (Wackett et al., 1989).

Propane induced isolates, *M. vaccae* JOB5 and *Rhodococcus jostii* RHA1, degraded 1,4-dioxane (Hand et al., 2015). Propanotrophs, along with propane bioparging, have also been successfully used for *in situ* site remediation (Bell et al., 2022; Divine et al., 2024; Horst et al., 2019; Lippincott et al., 2015). A key functional enzyme for propane oxidation is propane monoxygenase. This is coded by four subunits: propane monoxygenase large subunit (*prmA*), propane monoxygenase reductase subunit (*prmB*), propane monoxygenase small subunit (*prmC*) and propane monoxygenase coupling protein (*prmD*) (Kotani et al., 2003; Sharp et al., 2007). These enzymes have been reported within both the group 5 (Kotani et al., 2003) and group 6 (Deng et al., 2018a; Kotani et al., 2006; Masuda et al., 2012) soluble di-iron monoxygenases (SDIMO) (Yang et al., 2024). Previous research has directly linked both group 5 and 6 propane monoxygenases to contaminant biodegradation (Chen et al., 2023; Deng et al., 2018a; Lippincott et al., 2015; Sharp et al., 2007). Group 5 propane monoxygenase were also detected in 1, 4-dioxane degrading mixed cultures in our previous work (Eshghdoostkhatami et al., 2025).

In this study, we investigated the capacity of propane-enriched microbial communities to cometabolically degrade three chlorinated ethenes individually (TCE, cDCE and 1,1-DCE) under aerobic conditions. 1,1-DCE biodegradation is of particular interest because others have noted greater cytotoxic effects from 1,1-DCE degradation compared to

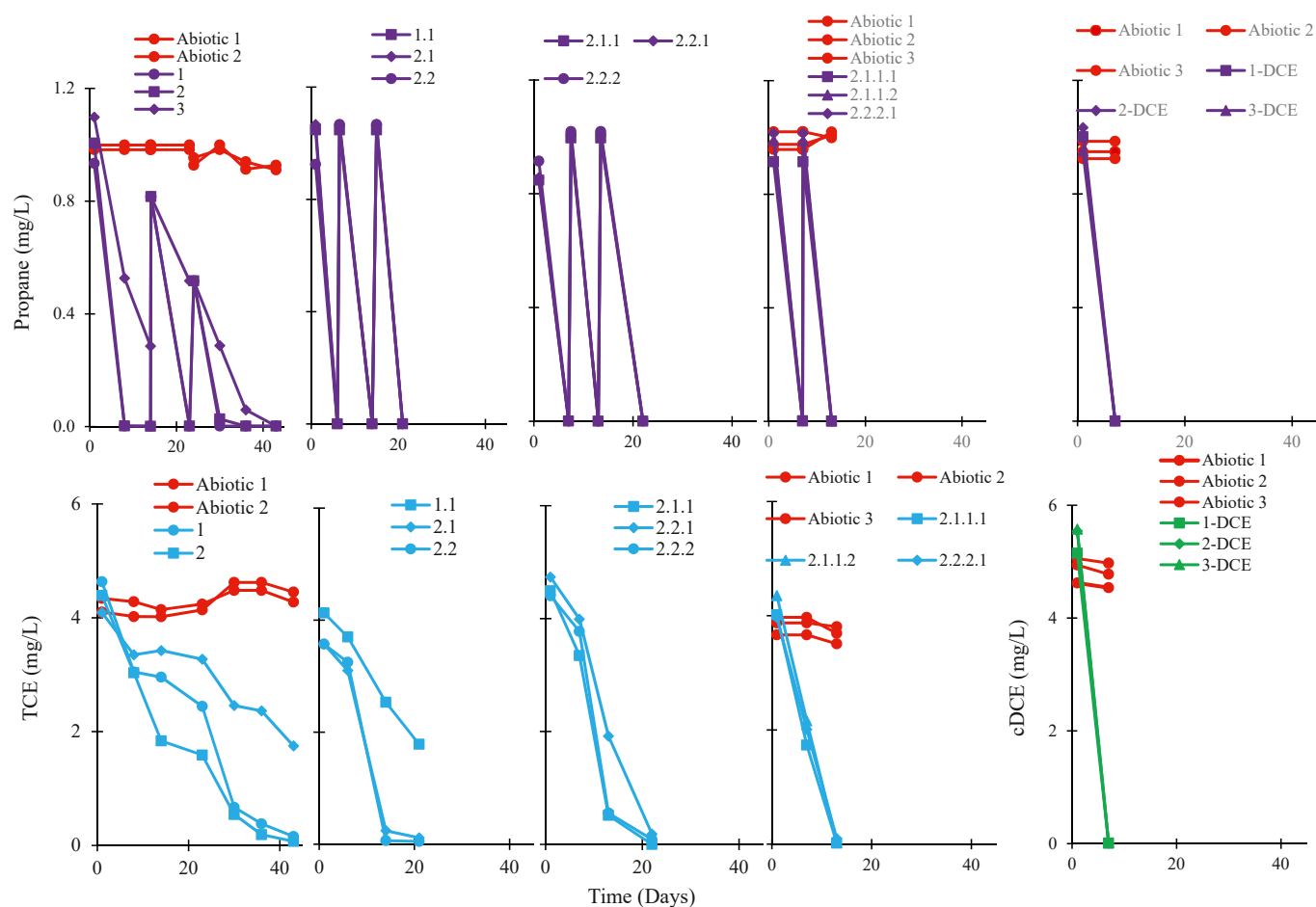


Fig. 1. Propane (top graphs) and TCE (bottom left four graphs) concentrations in replicate transfer propanotrophic microcosms and abiotic controls inoculated with agricultural soil T2. The transfer ratio each time was 10 mL culture in 20 mL media. Following the final transfer, the microcosms were amended with cDCE (bottom right).

other chlorinated ethenes (Dolan and McCarty, 1995; Ewers et al., 1990; Kim et al., 2000). Toxicity from 1,1-DCE transformation was also thought to be responsible for a stall in the cometabolic transformation of cDCE, 1,4-dioxane and 1,1,1-TCA in groundwater microcosms (Rolston et al., 2022) and for the limited decline in cDCE, 1,1-DCA and 1,4-dioxane at a field site (Danko et al., 2023). The propanotrophic enrichment cultures were derived from agricultural soils and impacted site sediments in a previous study focused on 1,4-dioxane biodegradation (Eshghdoostkhatami et al., 2025). Here, the cultures were evaluated for chlorinated ethene biodegradation across successive transfers to assess both transformation potential and stability. To link the observed activity with microbial function, whole genome sequencing was used to reconstruct community composition and recover metagenome-assembled genomes (MAGs), with a focus on identifying propane monoxygenase operons and associated metabolic features.

The specific objectives of the current study were to 1) determine the sustainability of chlorinated ethene biodegradation in propanotrophic mixed communities, 2) determine which propanotrophs survive well when exposed to individual chlorinated ethenes, 3) retrieve MAGs for dominant propanotrophs and obtain molecular data for propane monoxygenase operons and surrounding genes, and 4) compare the occurrence of group 5 versus putative group 6 propane monoxygenases.

2. Methods

2.1. Chemicals and source of cultures

Unlabeled TCE was purchased from Sigma-Aldrich (St. Louis, MO, USA) and propane gas was obtained from CAL GAS Direct (CA, USA). The propanotrophic enrichment cultures were developed from a previous study examining the biodegradation of 1,4-dioxane using inocula from various sources (three agricultural soils, impacted site sediment) (Eshghdoostkhatami et al., 2025). The three agricultural soil samples were from the Michigan State University (MSU) Main Cropping System Experiment at the Kellogg Biological Station Long-Term Ecological Research (KBS LTER) in southwest Michigan. Additional details of the agricultural soils can be found at <https://lter.kbs.msu.edu/research/long-term-experiments/main-cropping-system-experiment/>. Herein, the agricultural soils will be called Soil T2, Soil T3 and Soil T4. The site sediment was obtained from a West Coast Naval Station in California (herein called Site 1). Propanotrophic enrichment cultures were established through repeated cycles of propane amendments prior to the start of this study (Eshghdoostkhatami et al., 2025). These pre-enriched cultures were subsequently used as inocula for the experiments described here. Additional details on the development of the propanotrophic cultures have previously been provided (Eshghdoostkhatami et al., 2025).

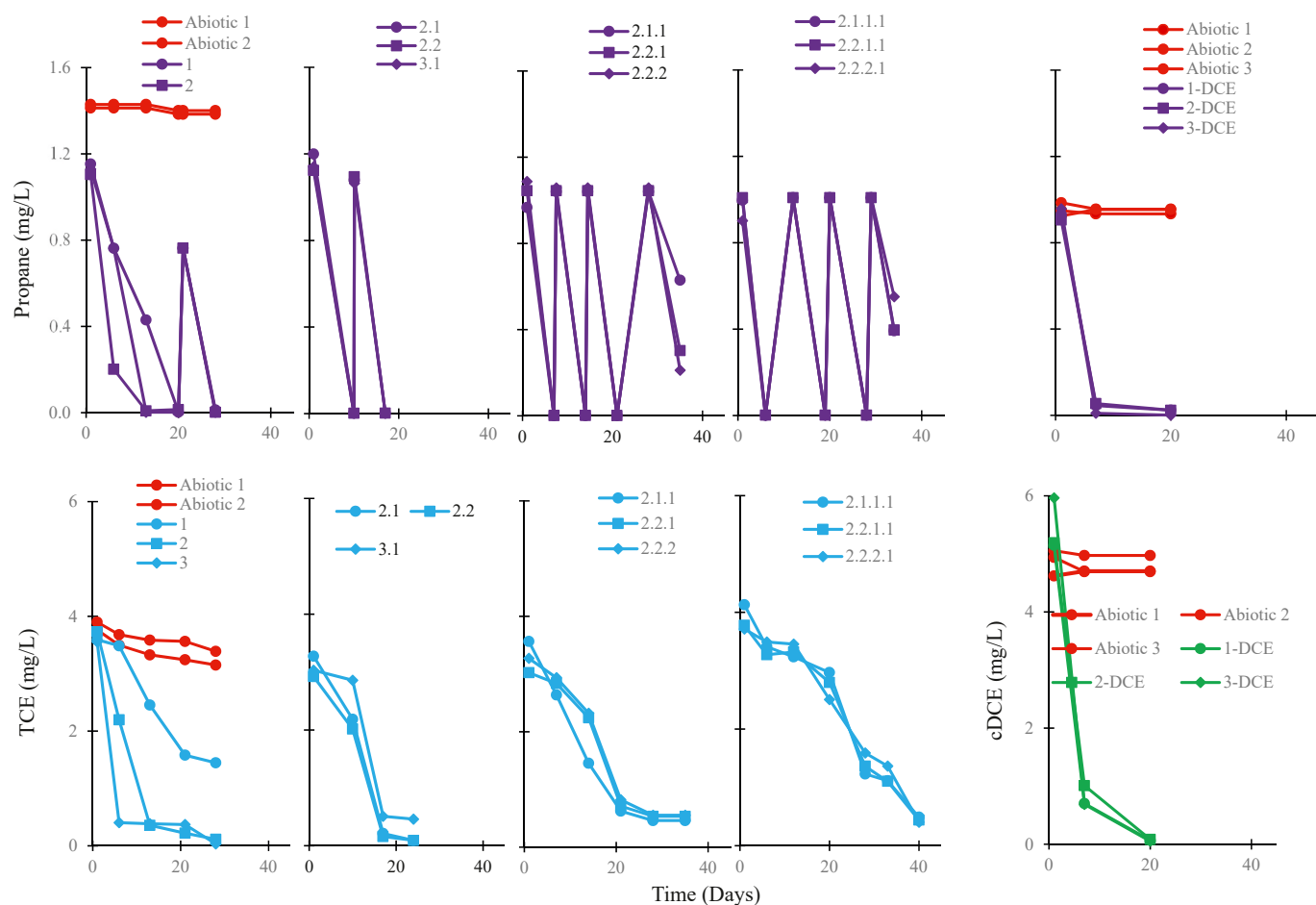


Fig. 2. Propane (top graphs) and TCE (bottom left four graphs) concentrations in replicate transfer propanotrophic microcosms and abiotic controls inoculated with agricultural soil T3. The transfer ratio each time was 10 mL culture in 20 mL media. Following the final transfer, the microcosms were amended with cDCE (bottom right).

2.2. Microcosm preparation and incubations

The initial microcosms for this study, from (Eshghdoostkhatami et al., 2025), consisted of 160 mL serum bottles containing 10 g (wet weight) of soil or sediment and 30 mL of liquid media, with the remaining volume serving as the gas headspace. Bottles were sealed with 20 mm butyl rubber stoppers (Fisher Scientific, product no. 50143854) and aluminum crimp seals (MilliporeSigma, Merck KGaA). All microcosms were incubated at room temperature on an orbital shaker operated at 120 rpm. The liquid medium was prepared using a buffer stock solution and a trace metal stock solution. The buffer stock consisted of K_2HPO_4 (32.4 g/L), KH_2PO_4 (10 g/L), and NH_4Cl (1 g/L), while the trace metal stock contained $MgSO_4 \cdot 7H_2O$ (2 g/L), $FeSO_4 \cdot 7H_2O$ (0.12 g/L), $MnSO_4 \cdot H_2O$ (0.03 g/L), $ZnSO_4 \cdot 7H_2O$ (0.03 g/L), and $CoCl_2 \cdot 6H_2O$ (0.01 g/L), as previously described (Faghihinezhad et al., 2025). Growth media was prepared by adding 100 mL of each stock solution to 800 mL of deionized water. The solution pH was then adjusted to 7.4 and placed in the autoclave for sterilization. After sterilization, the media were sonicated for 15–20 min in an ultrasonic bath to dissolve any precipitated salts.

For each inocula type (Soil T2, SoilT3, SoilT4 and Site1) three to four replicate microcosms were prepared along with corresponding abiotic controls. The chlorinated ethenes, TCE (4 to 5 mg/L), cDCE (4 to 5 mg/L) and 1,1-DCE (0.5 to 1 mg/L) were added to the active microcosms and abiotic controls. To protect against nutrient depletion and the accumulation of metabolic byproducts, serial transfers were performed for each microcosm, with up to five successive transfers being conducted over the

incubation period (if biodegradation was observed). Except in three cases (at the start of the experiment) the majority of serial transfers included 10 mL culture into 20 mL fresh media. The dilution ratio was changed (from 1:9 to 1:2) to favor the sustained biodegradation of each chemical and protect against losing any less abundant, yet active microorganisms. Following the degradation of cDCE and 1,1-DCE over multiple transfers, the cultures were then tested for their ability to degrade TCE. An additional set of experiments involved testing the cDCE and TCE degrading cultures for 1,1-DCE degradation (all approx. 4–5 mg/L). In another set of experiments, after the degradation of TCE over multiple transfers, the cultures were tested for their ability to degrade cDCE. As 1,1-DCE was only degraded by a subset of the cultures (at low concentrations), only two sets of microcosms and controls were then tested for TCE biodegradation potential.

2.3. Analytical methods

TCE, cDCE, and 1,1-DCE concentrations were determined by analyzing headspace samples with a gas chromatograph equipped with an electron capture detector (GC-ECD; PerkinElmer AutoSystem). Gas-phase samples were withdrawn using a 250 μ L gastight syringe (VICI Precision Sampling, Baton Rouge, LA, USA) and injected onto an AT-624 capillary column (30 m length, 0.53 mm internal diameter, 3.0 μ m film thickness). The detector temperature was maintained at 350 °C. The oven temperature program ramped from 50 °C to 100 °C at 10 °C min⁻¹, reaching the final temperature within 5 min for each analysis. The gas concentrations derived from the GC were converted to liquid

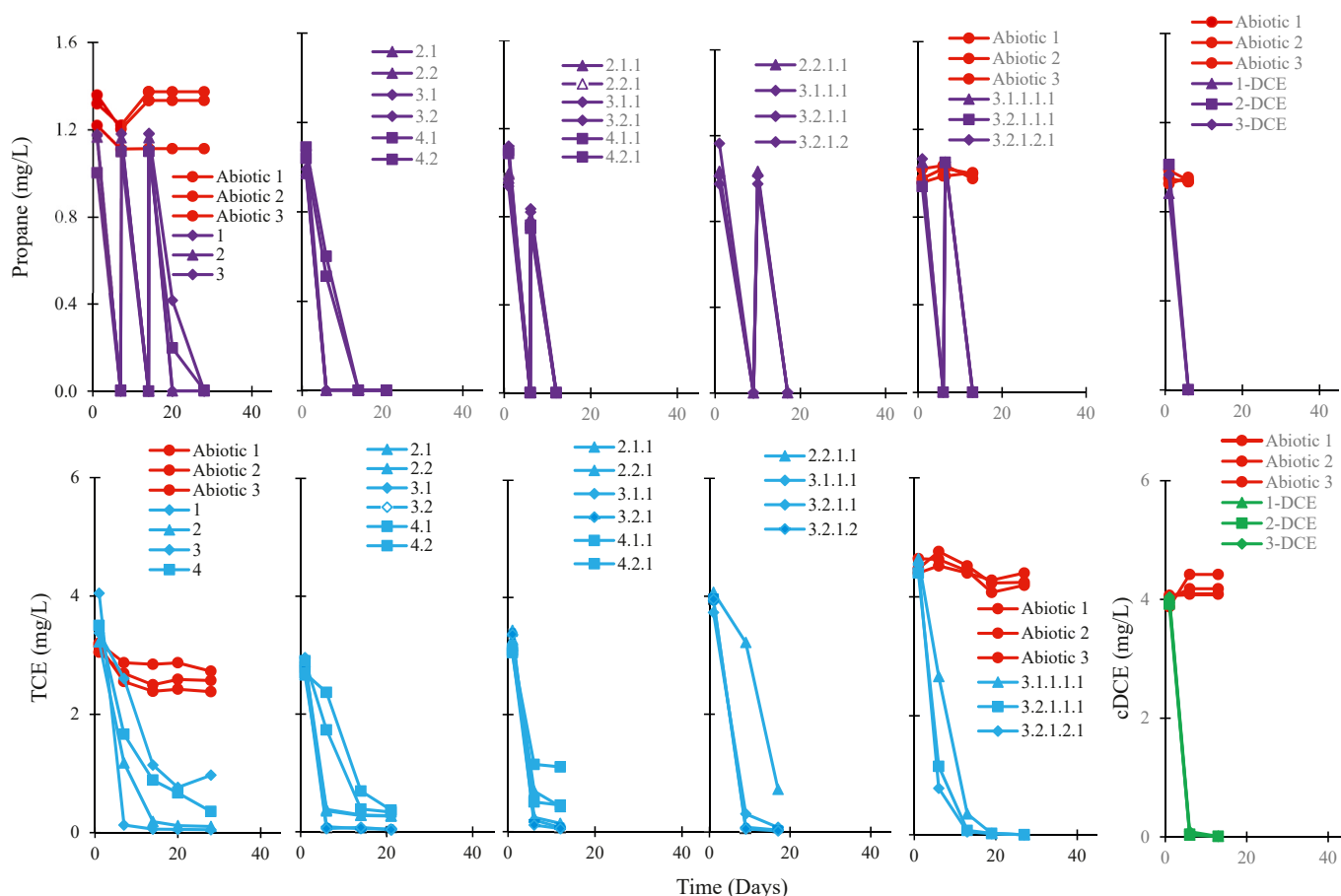


Fig. 3. Propane (top graphs) and TCE (bottom left four graphs) concentrations in replicate transfer propanotrophic microcosms and abiotic controls inoculated with agricultural soil T4. The transfer ratio each time was 10 mL culture in 20 mL media (except for the first transfer, which was 3 mL culture in 27 mL media). Following the final transfer, the microcosms were amended with cDCE (bottom right).

concentrations using Henry's Law. Henry's Law constants at 21 °C for TCE, cDCE, and 1,1-DCE were 0.3165, 0.1289, and 0.935 (EPA, 2026; Gossett, 1987). Propane concentrations were measured using the same gas chromatographic system and column configuration, operated with a flame ionization detector (GC-FID). For propane analyses, the oven temperature was held isothermally at 45 °C throughout each run. The pH of the liquid media was measured using a benchtop pH meter (Thermo Orion 5-Star, USA).

2.4. Genomic sequencing and analysis

Twenty DNA extracts were submitted for library generation and Whole Genome Sequencing (WGS) to the RTSF Genomics Core at MSU. This included eight DNA extracts that were amended repeatedly with TCE (replicates for Site 1, Soils T2, T3 and T4), eight that were amended repeatedly with cDCE (replicates for Site 1, Soils T2, T3 and T4), and four that were repeatedly amended with 1,1-DCE (replicates for Soils T2 and T3). DNA was extracted using the DNeasy PowerLyzer PowerSoil Kit (Mo Bio, USA) according to the manual protocol, following the last amendment of TCE, cDCE, and 1,1-DCE. The concentration of DNA in each extract was quantified using the Quant-iT™ dsDNA High-Sensitivity Assay Kit.

Libraries were prepared using the Quantabio sparQ DNA Library Preparation Kit with IDT xGen Unique Dual Index adapters following manufacturer's recommendations. Completed libraries were QC'd and quantified using a combination of Biotium AccuGreen High Sensitivity dsDNA and Agilent 4200 TapeStation HS DNA1000 assays. All libraries were normalized to a constant molar concentration prior to pooling.

Libraries were pooled in equimolar amounts and the pool quantified using the Invitrogen Colibri Quantification qPCR kit. The pool was sequenced using an Element AVITI Cloudbreak Freestyle High Output 300 cycle kit. Sequencing was performed in a 2x150bp paired end format. Base calling was done by AVITIOS v3.4.1 and the output was demultiplexed and converted to FastQ format using Element Biosciences bases2fastq v2.2.0.

The shotgun sequencing files were uploaded using Globus to KBase (Arkin et al., 2018). The data sets were subject to filtering and quality control using FastQC (version 0.12.1) (Andrews, 2010) and Trimmomatic (version 0.36) (sliding window size equal to 4, sliding window minimum quality equal to 15) (Bolger et al., 2014). The reads were subject to classification using Classify Taxonomy of Metagenomic Reads with Kaiju (version 1.9.0.) (Menzel et al., 2016).

Following this, Megahit was used to assemble the sequences (version 1.2.9) (Li et al., 2015) with a minimum contig length of 2000 bp. The assemblies were annotated with RASTtk (version 1.073) (Aziz et al., 2008) and Prokka (version 1.14.16) (Seemann, 2014) and the annotated assemblies were downloaded for collection of the targeted sequence subunits and gene positions on the appropriate contigs. A subset of the nucleic acid sequences (*prmA*) were subject to blastn (Altschul et al., 1990) on the NCBI website for the creation of phylogenetic trees (as described below). Also, after quality control, the assembled genomes were classified using Classify Taxonomy of Metagenomic Reads with Kaiju (version 1.9.0) (Menzel et al., 2016). Additionally, the contigs (from Megahit) were binned using CONCOCT (version 1.1) (Alneberg et al., 2014), MaxBin2 (version 2.2.4) (Wu et al., 2016) and MetaBAT2 (version 1.7) (Kang et al., 2015). The app Das Tool (version 1.1.2)

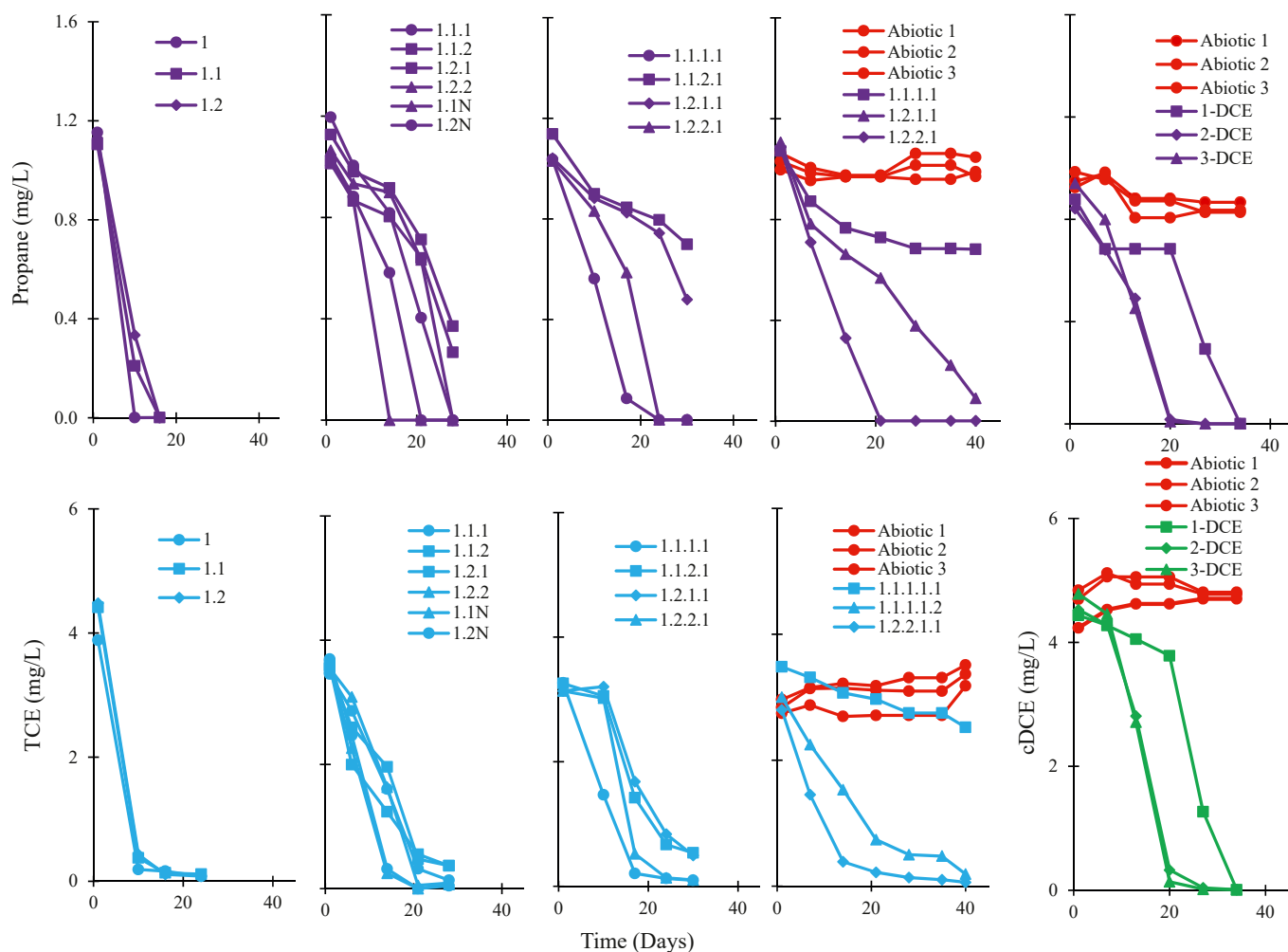


Fig. 4. Propane (top graphs) and TCE (bottom left four graphs) concentrations in replicate transfer microcosms and abiotic controls inoculated with sediment from an impacted site. The transfer ratio each time was 10 mL culture in 20 mL media (except for the first transfer, which was 3 mL culture in 27 mL media). Following the fourth transfer, the microcosms were amended with cDCE (bottom right).

(Sieber et al., 2018) was then used to optimize the binned contigs and CheckM (version 1.0.18) was used to quality filter the bins (80% completeness, 20% contamination) (Parks et al., 2015). The quality filtered bins were extracted using Extract Bins as Assemblies from BinnedContigs (version 1.0.2) (Arkin et al., 2018). Finally, the extracted bins were annotated using Annotate Metagenome Assembly with Prokka (version 1.14.5) (Seemann, 2014) and the taxonomy of each extracted bin was determined using Classify Microbes with GTDB-Tk (version 1.7.0) (Chaumeil et al., 2019). The relative abundances of.

MAGs containing the propane monooxygenase operon were estimated by aligning QC-filtered reads to individual MAGs using Bowtie2 (version 2.3.2) (Langmead and Salzberg, 2012). The assembly statistics and quality of propanotrophic MAGs were determined using Summarize GenomeSet (version 1.8.0) (Chivian et al., 2023; Karaoz and Brodie, 2022; Parks et al., 2015). The assembly statistics (e.g. genome length, completeness, contamination levels) of the propanotrophic MAGs are summarized in Table 1 and Supplementary Table 1.

Phylogenetic trees using nucleic acid sequences for *prmA* were generated using MEGA 12 (Kumar et al., 2024). For this, each *prmA* sequence was uploaded to NCBI for a nucleotide-nucleotide blastn search to find highly similar sequences (Altschul et al., 1990). Then, the sequences were downloaded and aligned with the sequences from the current study, using ClustalW with the default parameters (Thompson et al., 1994). The trees also contained *prmA* and *thmA* sequences from previously reported propane monooxygenase operons. The Maximum

Likelihood method with 1000 bootstrap replications was used to create the trees. The orientation and size of all subunits in the propane monooxygenase operons were shown in gene arrow plots created with R (Version 4.2.1) (R Core Team, 2018) in RStudio (Version, 2022.12.0) (RStudio Team, 2020) and the R packages ggplot2 (Version 3.4.4) (Wickham, 2016), readxl (Version 1.4.2) (Wickham and Bryan, 2023) and gggenes (Version 0.5.1.) (Wilkins, 2023). The output of Kaiju was used to create two phylogeny relative abundance plots using ggplot2 (Version 3.4.4) (Wickham, 2016) and readxl (Version 1.4.2) (Wickham and Bryan, 2023). The raw sequences were submitted to NCBI under Bioproject ID PRJNA1419878 (BioSample accession numbers SRR37150467 - SRR37150486). The quality controlled reads, MAGs (fasta assemblies) and the prokka annotations for each assembly are available in the following publicly available KBase narratives: TCE Site 1A and B Propanotrophic MAGs, TCE Soil 2A and B Propanotrophic MAGs, TCE Soil T3 A and B Propanotrophic MAGs, TCE Soil T4 A and B Propanotrophic MAGs, cDCE Site 1A 1B Propanotrophic MAGs, cDCE Soils T2 A and B Propanotrophic MAGs, cDCE Soil T3 A and B Propanotrophic MAGs, cDCE Soil 4A and B Propanotrophic MAGs, 1,1-DCE T2 T3 Propanotrophic MAGs.

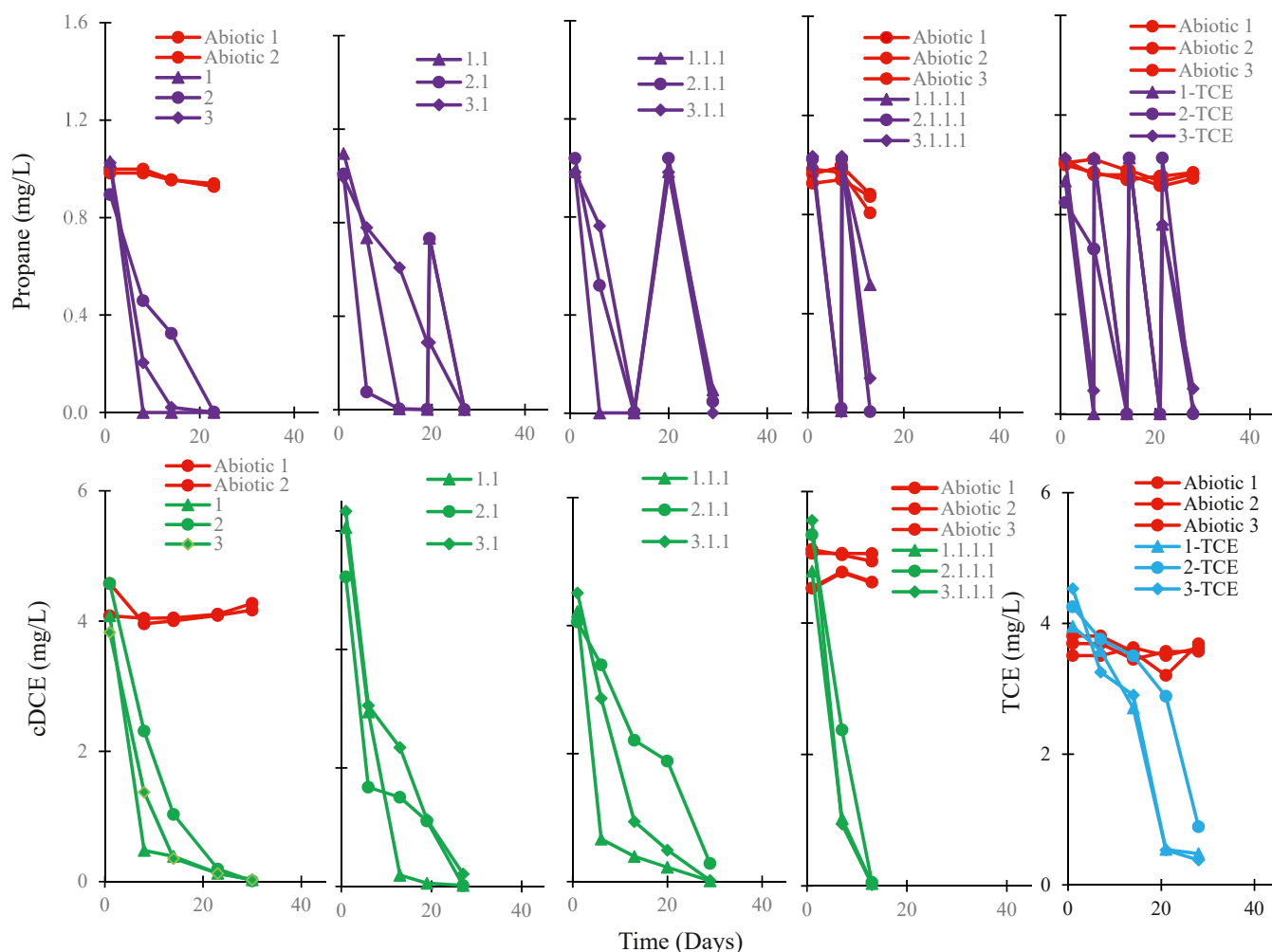


Fig. 5. Propane (top graphs) and cDCE (bottom left four graphs) concentrations in replicate transfer propanotrophic microcosms and abiotic controls inoculated with agricultural soil T2. The transfer ratio each time was 10 mL culture in 20 mL media. Following the final transfer, the microcosms were amended with TCE (bottom right).

3. Results

3.1. Source of propanotrophic cultures

As stated above, the propanotrophic enrichment cultures were generated in a previous study focusing on effective 1,4-dioxane biodegradation (Eshghdoostkhatami et al., 2025). 1,4-Dioxane (~3.5 mg/L) was degraded in the cultures used in the current study from between 7 and 18 days (Supplementary Fig. 1). Replicate cultures exhibited 1,4-dioxane biodegradation to below the 1,4-dioxane detection level of 0.46 µg/L, as previously described (Eshghdoostkhatami et al., 2025). Following the establishment of stable propane-oxidizing activity in those microcosms, the same enrichment cultures were used to evaluate the degradation of TCE, cDCE, and 1,1-DCE, as described below.

3.2. TCE followed by cDCE biodegradation trends

Propanotrophic enrichment cultures, derived from soils T2, T3, and T4 as well as from sediment from an impacted site (Site 1), were evaluated for TCE biodegradation in the presence of propane (Figs. 1–4). During each transfer, TCE and propane were added simultaneously and serial dilutions were performed prior to subsequent TCE amendments. As propane was typically rapidly depleted, it was amended multiple times. At the beginning of each degradation cycle, TCE concentrations

ranged from approximately 4.0–4.5 mg/L, while propane concentrations ranged from approximately 0.8–1.2 mg/L (Figs. 1–4). In the abiotic controls, both propane and TCE concentrations remained relatively stable over time, indicating biological removal mechanisms. In the enrichment cultures derived from soils T2 and T4, TCE concentrations decreased following propane amendments across replicate microcosms and transfers (Figs. 1 and 3). In both sets of samples, complete TCE degradation required approximately 40 days in the first transfer. However, subsequent transfers resulted in more rapid TCE degradation (removal under 20 days), perhaps suggesting an enrichment of TCE degraders. In contrast, TCE removal trends for T3 enrichment cultures were initially fast, then slowed over subsequent transfers (Fig. 2). TCE removal trends in Site 1 enrichment cultures were similar to those observed for T3 enrichment cultures (Fig. 4).

Following the final transfer in the four sets of TCE-amended enrichment cultures, a sub-group of microcosms were amended with cDCE to evaluate their ability to transform an alternative chlorinated ethene under propane-amended conditions (Figs. 1–4, bottom right). cDCE removal was rapid in the T2 and T4 enrichment cultures and slower in the T3 and Site 1 enrichment cultures. Overall enrichment cultures from T2 and T4 illustrated enhanced removal trends for TCE and cDCE compared to those from T3 and Site 1. As indicated above, DNA was extracted for sequencing before cDCE was added.

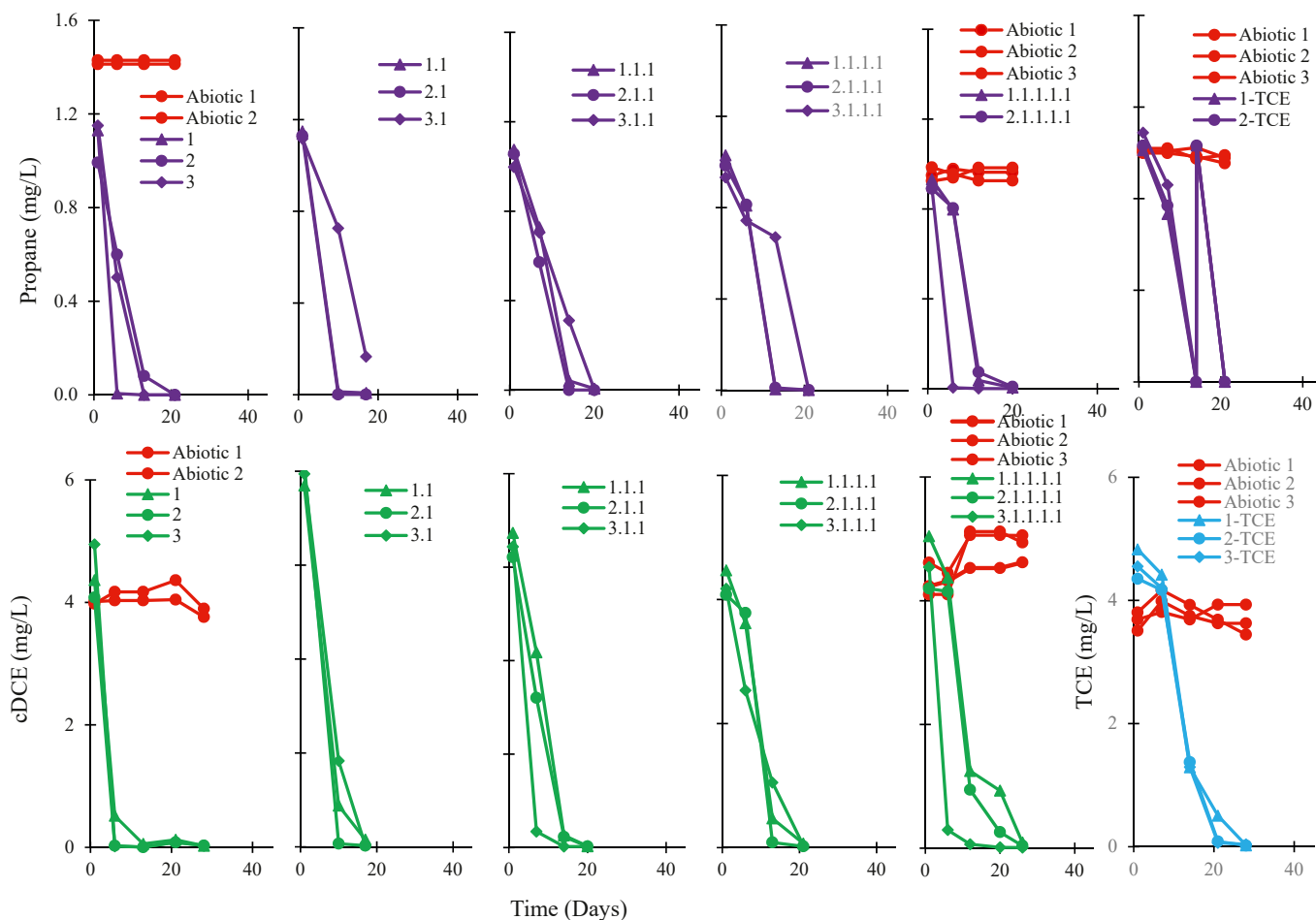


Fig. 6. Propane (top graphs) and cDCE (bottom left four graphs) concentrations in replicate transfer propanotrophic microcosms and abiotic controls inoculated with agricultural soil T3. The transfer ratio each time was 10 mL culture in 20 mL media. Following the final transfer, the microcosms were amended with TCE (bottom right).

3.3. cDCE followed by TCE biodegradation trends

Similar to the experiments described above, propanotrophic enrichment cultures, derived from the three soils and Site 1 were evaluated for cDCE biodegradation in the presence of propane, over successive transfers (Figs. 5–8). Initial cDCE and propane concentrations ranged between approximately 4–6 mg/L and 0.8–1.2 mg/L, respectively (Figs. 5–8). In the abiotic controls, both propane and cDCE concentrations remained relatively stable over time, indicating biological removal mechanisms.

In the propanotrophic enrichment cultures derived from soil T2, cDCE removal during the first transfer required 30 days. However, cDCE depletion in all replicates for the last transfer occurred in under 15 days, a trend suggestive of the enrichment of cDCE degraders (Fig. 5).

In general, in the soil T3 derived cultures, cDCE removal during each transfer was fast (under 20 days), although two replicates in the final transfer illustrated slower depletion (Fig. 6). In soil T4-derived enrichments, cDCE removal was relatively consistent between successive transfers and required approximately 30 days (Fig. 7). Site 1-derived propanotrophic enrichments also supported cDCE transformation during successive transfers with removal being observed in approximately 20 days (Fig. 8).

Following the final transfer in the cDCE-amended enrichment cultures, microcosms were amended with TCE to evaluate their capacity to transform this compound under propane-amended conditions (Figs. 5–8, bottom right). In contrast to the rapid cDCE degradation observed in transfer microcosms amended with TCE, TCE removal in these cultures

proceeded more slowly. The fastest removal (under 30 days) was noted in the propanotrophic enrichments derived from Soil T3 (Fig. 6). The slowest TCE removal (more than 40 days) was noted for the cultures derived from Site 1 (Fig. 8). For cultures derived from T2 and T4, TCE concentrations declined to approximately 0.5 mg/L within the monitored incubation period (25–28 days) (Figs. 5 and 7).

3.4. 1,1-DCE followed by TCE biodegradation trends

A subset of the cultures (all those amended with cDCE, then TCE) were tested for their ability to degrade 1,1-DCE at a concentration of 4–5 mg/L. In all cases, no decreases in 1,1-DCE concentrations were noted (within the approximately 30 day incubation period). Following this, all propanotrophic enrichment cultures previously amended with TCE or cDCE were tested for their ability to degrade 1,1-DCE at lower concentrations (between approximately 0.5 and 1.2 mg/L) (Fig. 9). Among those tested, all three replicates cultures derived from soils T2 and T3, previously amended with cDCE, exhibited both 1,1-DCE and propane biodegradation (Fig. 9). For the cultures previously amended with TCE, all three replicates derived from soil T3 exhibited 1,1-DCE biodegradation (Fig. 9, bottom right). However, propane degradation was only observed in one of the three replicates (Fig. 9, top right).

All additional experiments to test 1,1-DCE biodegradation only focused on propanotrophic enrichment cultures derived from T2 and T3, previously amended with cDCE (Figs. 10 and 11). For both sets of experiments, 1,1-DCE removal rates were initially fast (below 10 days), but slowed during successive transfers. In later transfers, 1,1-DCE

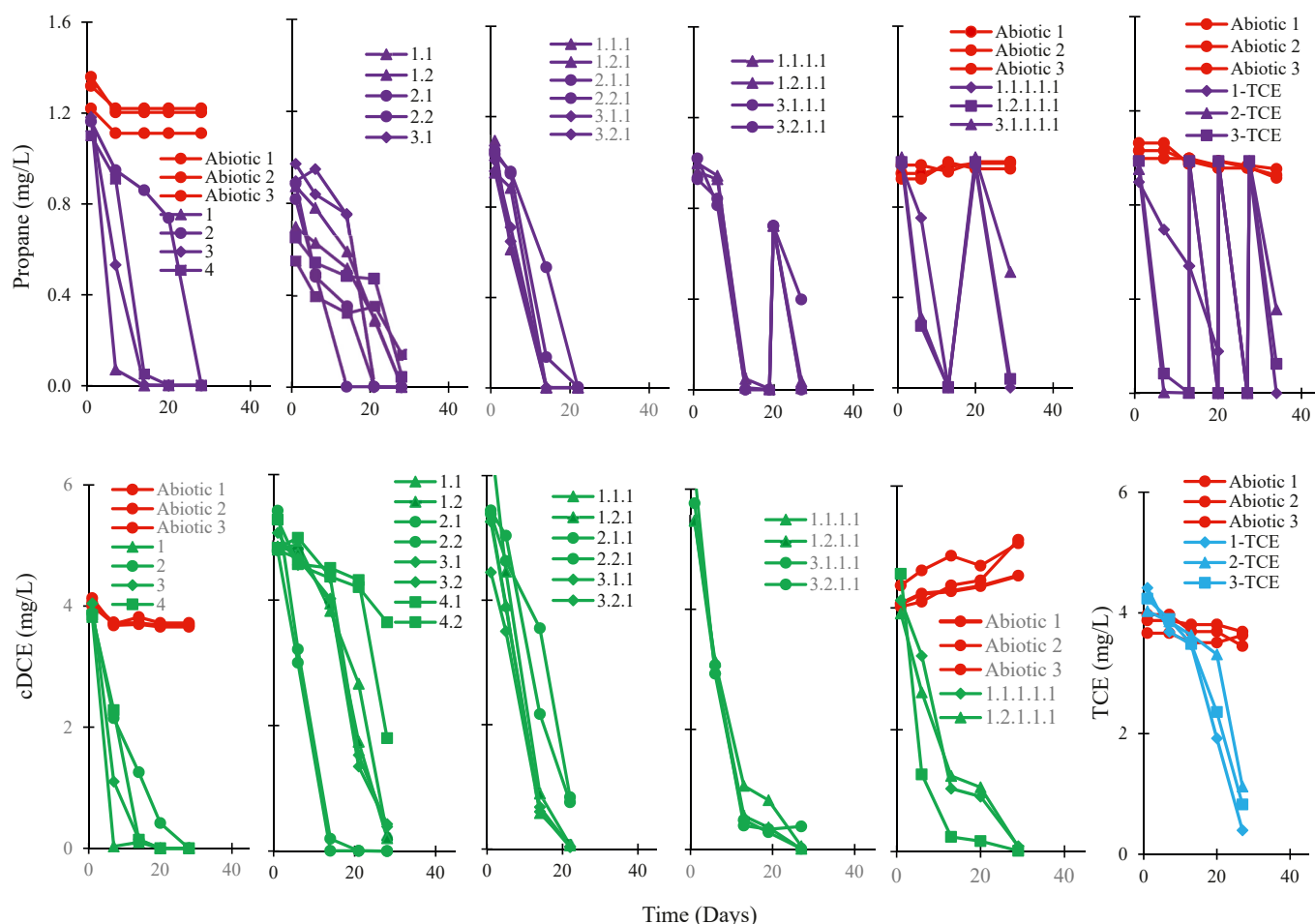


Fig. 7. Propane (top graphs) and cDCE (bottom left four graphs) concentrations in replicate transfer propanotrophic microcosms and abiotic controls inoculated with agricultural soil T4. The transfer ratio each time was 10 mL culture in 20 mL media (except for the first transfer, which was 3 mL culture in 27 mL media). Following the final transfer, the microcosms were amended with TCE (bottom right).

degradation required between approximately 18–25 days. Interestingly, propane biodegradation trends also slowed over successive transfers in the presence of 1,1-DCE. The second and third transfers for the enrichment cultures from soil T3 included additional microcosms with no propane (Fig. 11, bottom). Notably, for the third transfer without propane, 1,1-DCE removal was greatly reduced (compared to the microcosms with propane), indicating 1,1-DCE biodegradation was co-metabolically sustained on propane.

Similar to the cDCE experiments described above, the final step involved exposure to TCE (Figs. 10 and 11, bottom right). In soil T2-derived cultures, TCE degradation proceeded gradually and required approximately 50 days (Fig. 10). In soil T3-derived cultures, substantial decreases were observed over approximately 40 days before concentrations approached low levels, although one replicate stalled at ~50% TCE removal (Fig. 11). These incubation periods were notably longer than those observed for 1,1-DCE degradation in the same cultures (10–25 days across transfers). Also of interest is that during TCE degradation, propane biodegradation started again, suggesting propanotrophs were only temporarily inhibited in the presence of 1,1-DCE.

Overall, this subgroup of enrichment cultures first exhibited cDCE biodegradation, then 1,1-DCE biodegradation (both across multiple transfer steps), and ultimately TCE biodegradation, collectively indicating the capacity to transform all three chlorinated ethenes.

3.5. Microbial community composition

The taxonomic compositions of the propanotrophic enrichment

cultures were examined at the genus level using Kaiju on the reads (following quality control) (Fig. 12). The community structure differed between TCE- and cDCE-amended microcosms and also varied among inoculum sources. However, a key trend in the cDCE-amended enrichment cultures, from all four inoculum sources, was the dominance of the genus *Mycolicibacterium* followed by the genus *Mycobacterium* (Fig. 12, top). In contrast, in the TCE-amended enrichment cultures, all three enrichments from the agricultural soils (in particular those from soil T2) were dominated by the genus *Rhodococcus* (Fig. 12, bottom). The trend for the TCE-amended microcosms from the impacted site sediment differed from the others, being dominated by *Mycolicibacterium*, followed by *Mycobacterium* and *Rhodococcus*. In summary, the majority of the TCE amended cultures were dominated by *Rhodococcus*, whereas all cDCE amended cultures were dominated by *Mycolicibacterium* followed by the genus *Mycobacterium*.

The community composition was also examined in the two enrichment cultures capable of 1,1-DCE biodegradation (Supplementary Fig. 2). Both communities were originally amended with cDCE and were therefore dominated by *Mycolicibacterium* and *Mycobacterium* (as described above). However, following enrichment on 1,1-DCE, the community completely changed, with *Pseudonocardia* becoming the dominant genus in both replicates, for both soils. Other abundant genera included *Bradyrhizobium*, and *Mesorhizobium*. *Mycolicibacterium* and *Mycobacterium* were still present, but at lower relative abundance levels. Although *Pseudonocardia* MAGs were retrieved from each of the four samples, all contained a truncated large subunit for propane monooxygenase.

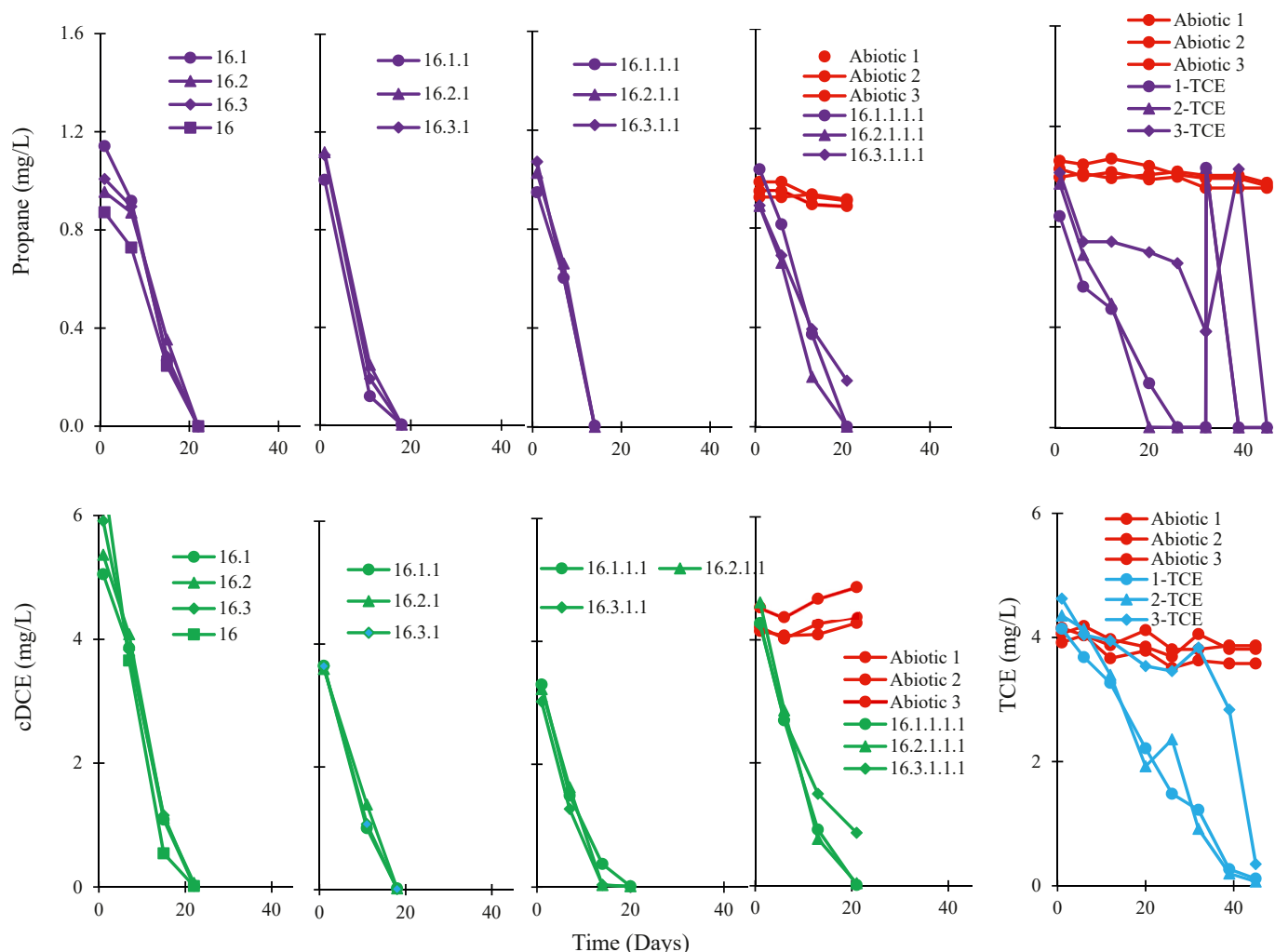


Fig. 8. Propane (top graphs) and cDCE (bottom left four graphs) concentrations in replicate transfer microcosms and abiotic controls inoculated with sediment from an impacted site. The transfer ratio each time was 10 mL culture in 20 mL media. Following the fourth transfer, the microcosms were amended with TCE (bottom right).

3.6. MAGs and phylotypes containing the propane monooxygenase operon

The majority of MAGs containing the propane monooxygenase operon, from all enrichment cultures, classified within the phylum *Actinobacteria* (*Rhodococcus*, *Mycobacterium* and *Pseudonocardia*) (Table 1). Two MAGs, from soil T3 and T4 cDCE amended enrichments, classified within the phylum *Proteobacteria* (*Methylibium* and an unclassified *Burkholderiaceae*). Consistent with the microbial community composition described above, the majority of MAGs from the cDCE amended cultures classified as *Mycobacterium*. In contrast, in the TCE amended enrichments, five classified as *Rhodococcus* (*R. opacus* or *R. wratislaviensis*), four classified as *Mycobacterium* and two classified as *Pseudonocardia broussonetiae* (only from the impacted site enrichments). Similar to the cDCE enrichments, the MAGs containing the propane monooxygenase operon (with subunits of the correct size) from the 1,1-DCE degrading cultures classified as *Mycobacterium*. The MAG relative abundance patterns (as implied by % of mapped reads, Table 1) were consistent with the microbial community composition described above. Specifically, *R. opacus* or *R. wratislaviensis* illustrated the highest relative abundance values in the TCE amended samples from T2, T3 and T4. Similarly, *Mycobacterium* MAGs illustrated the highest relative abundance values in the cDCE amended samples.

Eight (from nineteen) and one (from twelve) MAGs from the cDCE

and TCE amended cultures, respectively, contained propane monooxygenase operons with different order and annotation patterns from the rest. All, except one, were associated with *Mycobacterium* MAGs. The large subunit was annotated as methane monooxygenase component A alpha chain (*mmoX*) and the effector was annotated as methane monooxygenase regulatory protein B (*mmoB*). Interestingly, these subunits are the same size as *prmA* and *prmD*. If these genes were indeed *prmA* and *prmD*, the order would be *prmACDB*, an order previously been reported for propane monooxygenase operons in other *Mycobacterium* strains (*M. chubuense* strain NBB4, *M. dioxanotrophicus* strain PH-06, *Mycobacterium* sp. TY-6, *M. vaccae* JOB5) (Chen et al., 2023; Cupples, 2024; He et al., 2017a; Kotani et al., 2006). One hypothesis, based on the subunit order and size, is that these operons represent propane monooxygenases classifying as group-6 SDIMOs. Based on the MAG assembly statistics (Table 1 and Supplementary Table 1), the majority (28 from 33) of the propane monooxygenase containing MAGs were classified as “near complete”, as previously defined (completeness >90%, contamination ≤5%, rRNA/tRNA requirement not met (Bowers et al., 2017; Cansdale and Chong, 2024). No low quality MAGs were included.

Propane monooxygenase operons, absent from any MAG, were also obtained from the culture metagenomes (Table 2). The closest match for these operons in GenBank included those in the genera *Mesorhizobium*, *Nocardioides*, *Methylibium*, *Rhodococcus* and *Mycobacterium*. Four operons illustrated the expected subunit sequence and annotation

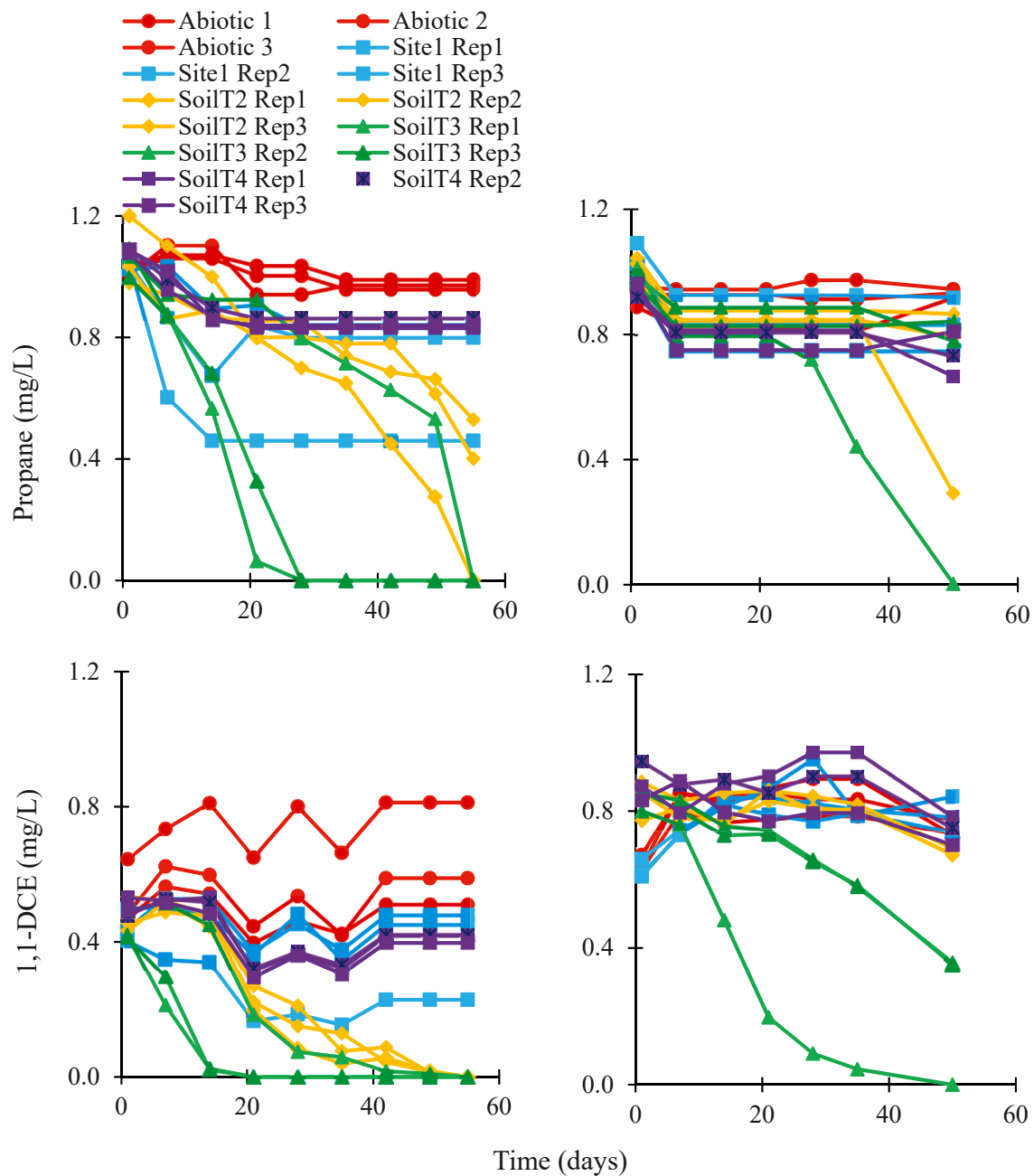


Fig. 9. Propane and 1,1-DCE concentrations in microcosms originally amended with cDCE (left) and TCE (right) in replicate microcosms derived from soils and Site 1 sediment.

(*prmABCD*). Six operons exhibited the same pattern discussed above (*prmACDB*), with the closest matches in GenBank being *Rhodococcus* or *Mycobacterium*.

3.7. Propane monooxygenase operons (*prmABCD*) and flanking genes

The size and orientation of the four subunits of the propane monooxygenase operons, as well as the flanking genes, have been summarized by phylotype (Figs. 13–15). The arrangement and order of these genes were the same for the three *R. opacus* MAGs (Fig. 13). Each contains two hypothetical genes directly downstream of *prmABCD*, followed by an alcohol dehydrogenase gene (*adhT*), a gene (*groL*) encoding for a molecular chaperone that assists protein folding (60 kDa chaperonin) (Frydman, 2001), followed by another hypothetical gene, and a gene (*COG0277*) encoding for a putative FAD-linked oxidoreductase. The

gene encoding for acetoin dehydrogenase operon transcriptional activator AcoR (*acoR*) is present on both sides of this group. The arrangement and order for the flanking genes from the two *R. wratislaviensis* were the same, yet differed from the pattern described for *R. opacus*. Specifically, two alcohol dehydrogenases are present downstream of *prmABCD* (*adh* and *adhT*), only one copy of *acoR* is present and *COG0277* is absent. All five *Rhodococcus* MAGs contain a gene encoding for catalase (*katA*) close to *prmABCD*, in the opposite direction. It is notable the patterns were the same for each species, yet different between them.

In contrast to the consistency in the arrangement of the flanking genes across the contigs from the *Rhodococcus* MAGs, the arrangement for the *Mycobacterium* MAGs varied (Fig. 14). In fact, eight different trends were observed, as illustrated by the numbers on the left of the figure. Common trends included the presence of *groL*, *adh* and/or *adhT* downstream of *prmABCD* (except in one case, number 2). One copy of

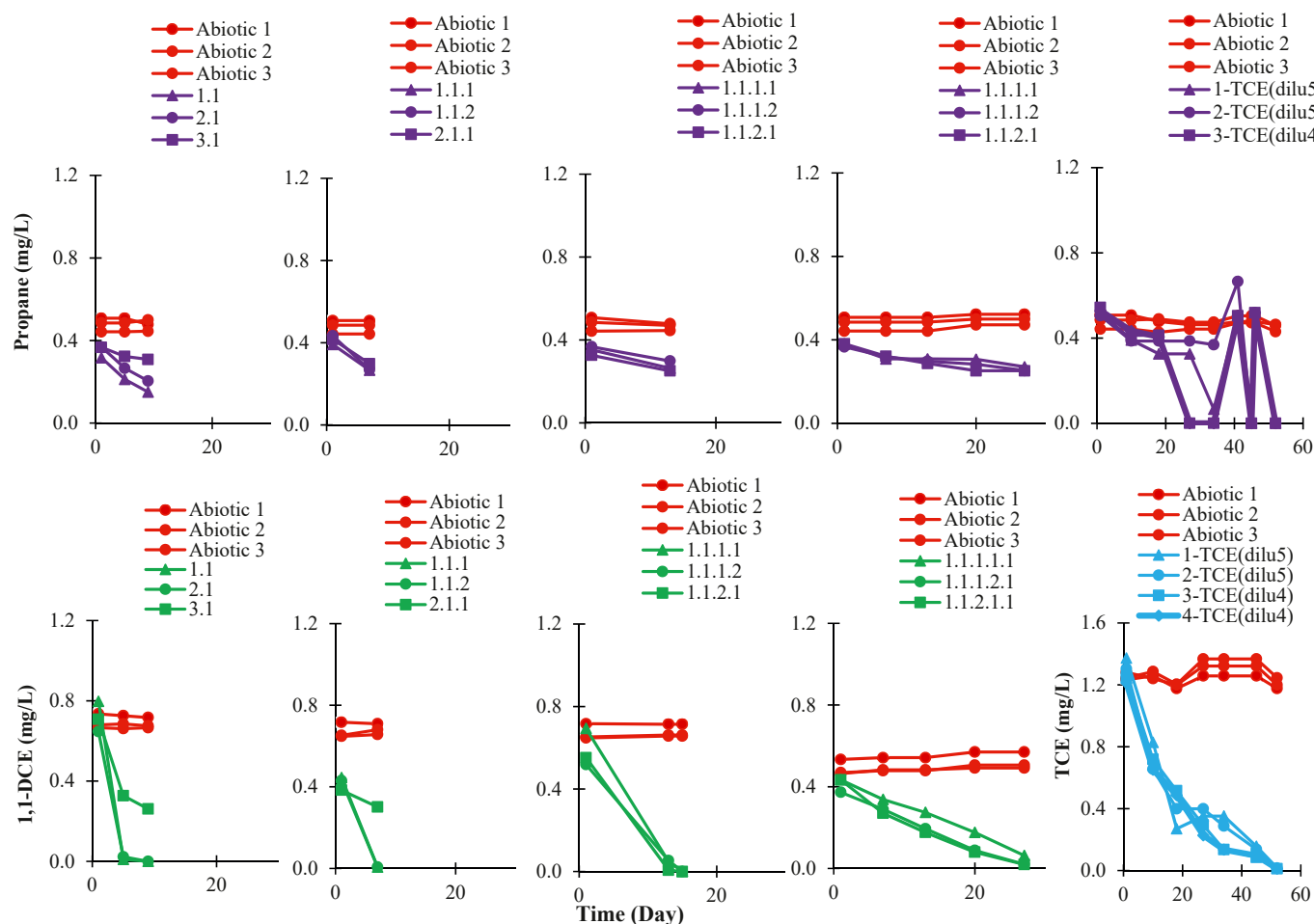


Fig. 10. Propane (top graphs) and cDCE (bottom left four graphs) concentrations in replicate transfer microcosms and abiotic controls inoculated with agricultural soil T2. The transfer ratio each time was 10 mL culture in 20 mL media. Following the fourth transfer, the microcosms were amended with TCE (bottom right).

acoR was noted in a subset of the *Mycobacterium* MAGs, and *COG0277* and *kata* were absent. A number of other genes, not observed in the *Rhodococcus* MAGs, were also present. As discussed above, the majority of the propane monooxygenase operon containing *Mycobacterium* MAGs were from the 1,1-DCE and cDCE amended cultures, with only four from the TCE amended cultures.

The flanking genes around *prmABCD* in metagenomes (most similar to those from the genera *Mesorhizobium*, *Methylibium* and *Nocardioides*) as well as MAGs classifying as *Methylibium* and *Pseudonocardia broussonetiae* were also determined (Fig. 15). In one case, for *Nocardioides*, the contig was short and limited data was obtained. For the others, consistent with the *Rhodococcus* and *Mycobacterium* MAGs, the majority contained *groL* downstream of *prmABCD*. The other five also contained *acoR*, however, the direction differed in two cases (*Methylibium* related sequences). Only the *Mesorhizobium* associated dataset contained an alcohol dehydrogenase gene.

3.8. Propane monooxygenase operons (*prmACDB*) and flanking genes

Flanking genes were also obtained for the propane monooxygenase operon *prmACDB* (possibly encoding for group-6 SDIMOs) (Supplementary Fig. 3–5). Two trends were noted for the *Rhodococcus* associated flanking genes upstream of *prmACDB* (Supplementary Fig. 3). However, all five contained *styD* (phenylacetaldehyde dehydrogenase), *adhD* (alcohol dehydrogenase) and *groL1* downstream of *prmACDB*. Unlike the *Rhodococcus* trends discussed above, *acoR* and *kata* were not detected. All *Mycobacterium* associated *prmACDB* contigs (primarily

from cDCE amended cultures) also contained *styD*, *adhD* and *groL1* (or *groL2*) immediately downstream (Supplementary Fig. 4). The upstream genes illustrated four different trends. Four *Pseudonocardia* MAGs contained this operon and similar flanking genes between the four, involving *groL*, hypothetical genes, *prop* (proline/betaine transporter) and *mntD* (manganese transport system membrane protein) (Supplementary Fig. 5). Genes encoding for alcohol dehydrogenases (*adh*, *adhT*, *adhD*) and *acoR* were absent close to *prmACDB*.

3.9. Phylogenetic analysis of *prmA* sequences

The group 5 *prmA* (propane monooxygenase large subunit) sequences were aligned to the closest matches in GenBank as well as *prmA* and *thmA* sequences from well-studied microorganisms (Figs. 16–18). The putative group 6 *prmA* sequences (from *prmACDB* operons) were aligned to those previously reported from both group 5 and group 6 operons (Fig. 19). The *Mycobacterium* *prmA* sequences largely clustered together and were similar to those, for example, from *Mycobacterium smegmatis*, *Mycobacterium goodii*, *Rhodococcus* sp. RR1, *Mycobacterium hodleri*, *Mycobacterium tokaiense*, *Mycobacterium murale* and *Mycobacterium littorale* (Fig. 16). The *prmA* sequences from *R. opacus* and *R. wratislaviensis* MAGs clustered with *prmA* sequences from the same species (Fig. 17). The *prmA* sequences from *Methylibium*, *Mesorhizobium*, *Pseudonocardia* are shown together (Fig. 18). Interestingly, the *Methylibium* associated *prmA* sequences from the current study clustered with those from *Methylibium petroleiphilum*, including the MTBE degrader *M. petroleiphilum* PM1 (Hanson et al., 1999). The

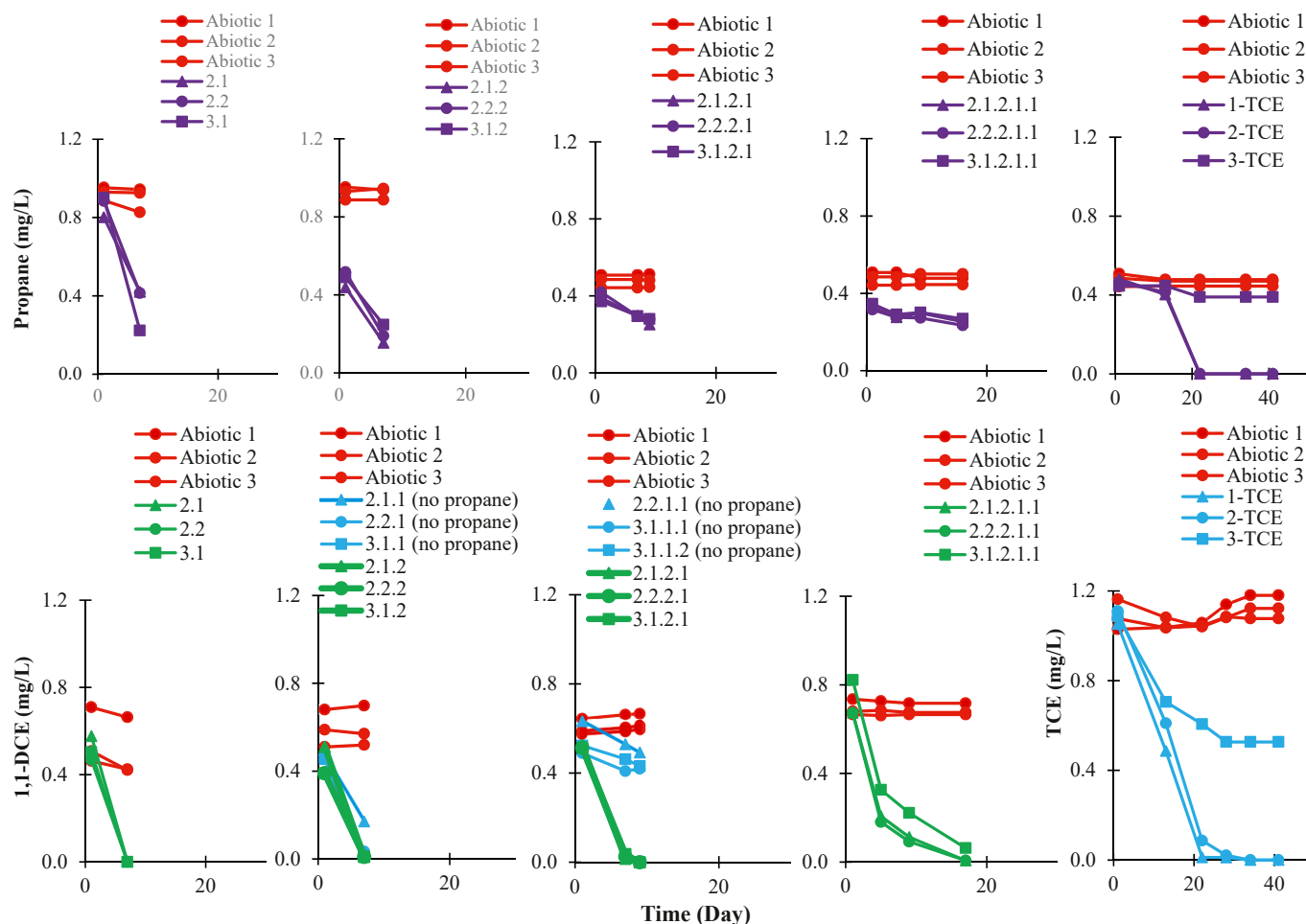


Fig. 11. Propane (top graphs) and 1,1-DCE (bottom left four graphs) concentrations in replicate transfer microcosms and abiotic controls inoculated with agricultural soil T3. The transfer ratio each time was 10 mL culture in 20 mL media. Following the fourth transfer, the microcosms were amended with TCE (bottom right).

Mesorhizobium associated sequences in the current study clustered with *prmA* sequences from this genus. The *P. broussonetiae* *prmA* sequences from the current study clustered with a *prmA* sequence from the same species and others from the same genus. The *Nocardioides* *prmA* sequence was most similar to that from *Rhodococcus* RR1. In general, the group 5 *prmA* sequences illustrated the closest similarity to those from *Rhodococcus* RR1 and *Gordonia* TY-7, with *thmA* and *prmA* (group 6) sequences forming separate branches (Figs. 16–18). As expected, the clustering patterns were notably different for the putative group 6 *prmA* sequences from the current study (Fig. 19). Specifically, the sequences most closely aligned with previously reported group 6 *prmA* sequences, including *M. chubuense* strain NBB4, *M. dioxanotrophicus* strain PH-06 and *R. wratislaviensis* IFP2016 (Auffret et al., 2015; He et al. 2017a, 2017b). In contrast, the group 5 *prmA* and *thmA* sequences formed separate branches (Fig. 19). These patterns provide additional evidence for the classification of these operons as group 6 SDIMOs.

4. Discussion

This study investigated the capacity of propane-enriched microbial communities to cometabolically transform three chlorinated ethenes (TCE, cDCE, 1,1-DCE) under aerobic conditions. The enrichment cultures demonstrated consistent biodegradation of TCE and cDCE over multiple transfers. TCE degrading cultures were capable of cDCE degradation and vice versa. In contrast, only a subset of the cultures degraded 1,1-DCE. It is important to note the chlorinated ethene degradation pathways were not investigated in this study. Future

research should examine the transformation products generated to ensure more toxic products are not produced.

Interestingly, metagenomic analysis indicated the cDCE degrading cultures were dominated by *Mycobacterium* and *Mycolicibacterium*, whereas the majority of the TCE degrading cultures were dominated by *Rhodococcus*. The occurrence of group 5 and putative group 6 propane monooxygenase operons in the cultures suggests their importance in mixed contaminant biodegradation (however their expression was not examined). The current study also provided novel information concerning the flanking genes of propane monooxygenase operons from MAGs classifying as *R. wratislaviensis*, *R. opacus*, *Mycobacterium*, *Methylobium* and *P. broussonetiae*. This project builds on our previous work documenting the occurrence of propanotrophic *Mycobacterium*, *R. opacus*, *R. wratislaviensis* MAGs in mixed cultures capable of biodegradation of 1,4-dioxane at both high and low concentrations (Eshghdoostkhatami et al., 2025).

The importance of propanotrophs for the biodegradation of environmental contaminants has previously been reported for a range of chemicals. For example, five *Mycobacterium* strains growing on propane degraded trichloroethene and *M. vaccae* JOB5 degraded cDCE, trans-dichloroethene (tDCE), 1,1-DCE, and vinyl chloride (Wackett et al., 1989) and TCE (Hand et al., 2015). Similarly, after growth on propane, several isolates (including *Mycobacterium vaccae* JOB5) degraded methyl tert-butyl ether (MTBE), ethyl tert-butyl ether and tert-amyl methyl ether (Steffan et al., 1997). The propanotrophs *Rhodococcus* sp. RR1 and *M. vaccae* JOB5 have been associated with the biodegradation of *n*-nitrosodimethylamine (Sharp et al., 2010). Further, propane grown

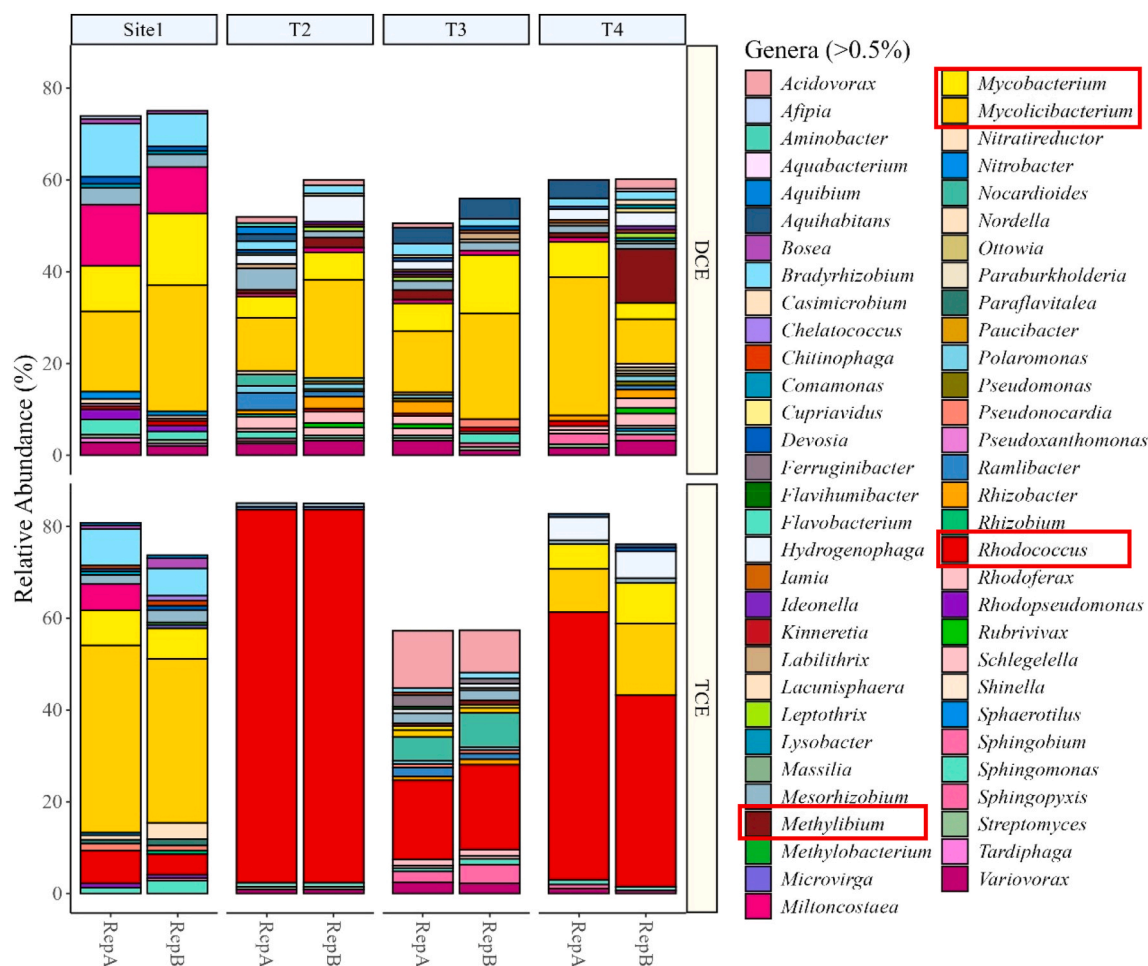


Fig. 12. Relative abundance (%) of genera (>0.5%) in replicate microcosms amended with trichloroethene (TCE, bottom) or *cis*-dichloroethene (cDCE, top). The microcosms were either inoculated with sediment from an impacted site (Site1) or agricultural soil from treatments 2 (T2), 3 (T3) or 4 (T4). Dominant genera are boxed in red.

Table 2

Summary of metagenomes that contained all four subunits of the propane monooxygenase (*prm*) operon. Some contained two subunits that were not annotated as *prm* *subunits (but were the correct length and orientation).

Substrate	Soil/Replicate	Contig	Closest Match in NCBI	<i>prm</i> ABCD	<i>prm</i> A*CD*B
cDCE	Site1B	k141_94871	<i>Mesorhizobium</i>	X	
cDCE	T2A	k141_101009	<i>Nocardioideis</i>	X	
cDCE	T2B	k141_181014	<i>Methylibium</i>	X	
TCE	T2A	k141_7491	<i>Rhodococcus</i>		X
TCE	T2B	k141_8699	<i>Rhodococcus</i>		X
TCE	T3B	k141_24812	<i>Rhodococcus</i>		X
TCE	T3B	k141_386592	<i>Rhodococcus</i>		X
cDCE	T4A	k141_73973	<i>Mycobacterium</i>		X
TCE	T4A	k141_15338	<i>Rhodococcus</i>		X
TCE	T4B	k141_207854	<i>Mycobacterium</i>	X	

Rhodococcus aetherivorans TPA co-metabolized 1,1,2,2-tetrachloroethane (Cappelletti et al., 2018), four propane oxidizing bacteria (*R. jostii* RHA1, *M. vaccae* JOB5, *Rhodococcus ruber* ENV425, *Sphingopyxis* sp. AX-A) degraded 1,2,3-trichloropropane (Wang and Chu, 2017) and propane induced isolates (*M. vaccae* JOB5 and *R. jostii* RHA1) degraded 1,4-dioxane (Hand et al., 2015). Propane amended mixed cultures have also been linked to the biodegradation of organic contaminants. Propane and inorganic nutrients stimulated 1,2-dibromoethane biodegradation in microcosms made with aquifer solids and groundwater (Hatzinger et al., 2015) and a propanotrophic enrichment culture derived from the site materials also degraded 1,2-dichloroethane (Hatzinger et al., 2015).

Here, the dominance of *Mycobacterium* in cDCE degrading cultures and *Rhodococcus* in TCE degrading cultures is particularly noteworthy. Consistent with this, TCE degradation has previously been associated with *Rhodococcus* strains, such as *Rhodococcus* sp. L4 (Suttinun et al., 2010), *Rhodococcus opacus* PD630 (Baskaran et al., 2020), *Rhodococcus rhodochrous* ATCC 21197 (Bealessio et al., 2023; Malachowsky et al., 1994), *Rhodococcus corallinus* B-276 (Saeki et al., 1999), and *Rhodococcus erythropolis* (Dabrock et al., 1992). Similarly, cDCE biodegradation has been previously reported for other *Mycobacterium* strains, such as *M. chubuense* NBB4 (Le and Coleman, 2011), *M. aurum* L1 (Hartmans and De Bont, 1992), and *Mycobacterium* sp. strain TRW-2 (Fathepure

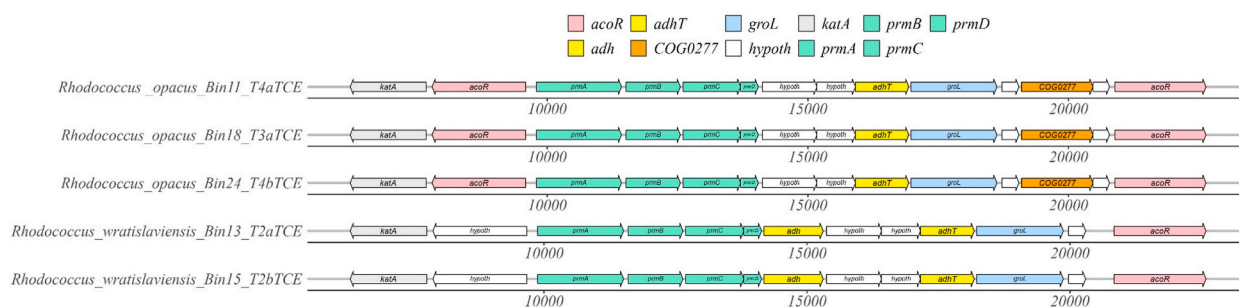


Fig. 13. Gene arrangement and length of the propane monoxygenase operon and surrounding genes from metagenome assembled genomes classifying as *Rhodococcus opacus* (top) and *Rhodococcus wratislaviensis* (bottom). All five were from TCE degrading microcosms. The encoded proteins are acetoin dehydrogenase operon transcriptional activator AcoR (*acoR*), alcohol dehydrogenases (*adh* and *adhT*), putative FAD-linked oxidoreductase (COG0277), 60 kDa chaperonin (*groL*), hypothetical proteins (*hypoth*), catalase (*katA*), propane monoxygenase large subunit (*prmA*), reductase component (*prmb*), small subunit (*prmc*) and effector component (*prmd*).

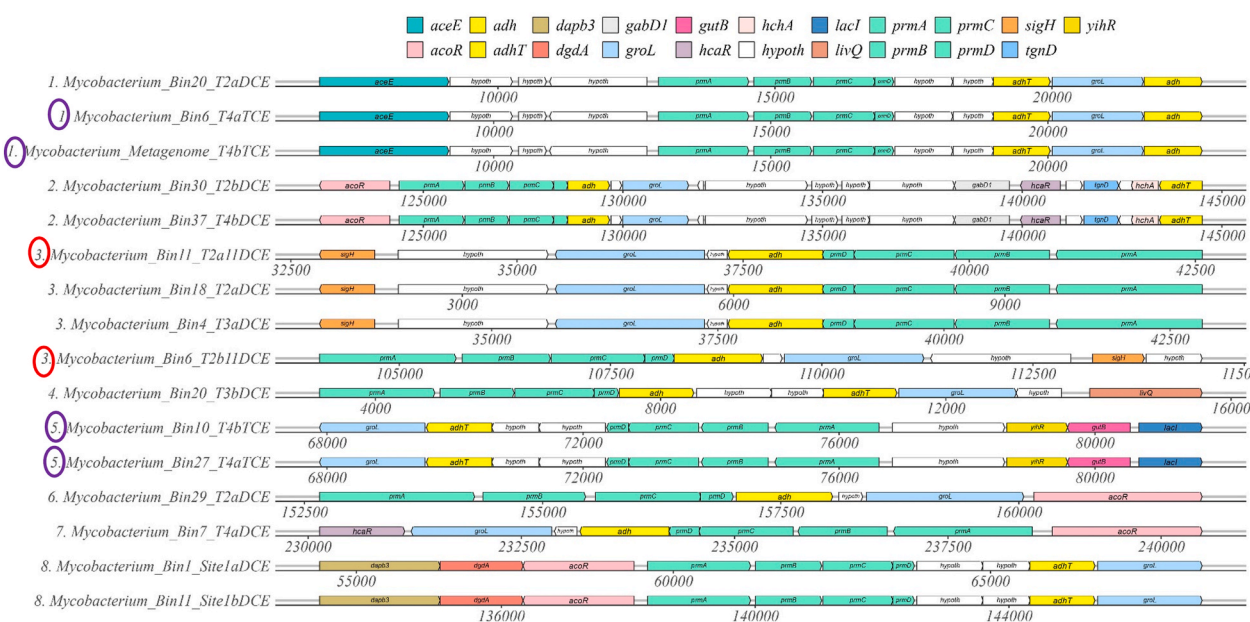


Fig. 14. Gene arrangement and length of the propane monoxygenase operon and surrounding genes from metagenome assembled genomes classifying as *Mycobacterium*. The numbers represent phylotypes with different trends for the surrounding genes. Those circled in red are from 1,1-DCE degrading microcosms and those circled in purple are from TCE degrading microcosms. All others are from cDCE degrading microcosms. The encoded proteins are pyruvate dehydrogenase E1 component (*aceE*), acetoin dehydrogenase operon transcriptional activator AcoR (*acoR*), alcohol dehydrogenase (*adh* and *adhT*), dipeptidyl aminopeptidase BIII (*dapb3*), 2,2-dialkylglycine decarboxylase (*dgdA*), succinate-semialdehyde dehydrogenase [NADP(+)] (*gabD1*), 60 kDa chaperonin (*groL*), sorbitol dehydrogenase (*gutB*), hca operon transcriptional activator HcaR (*hcaR*), protein/nucleic acid deglycase HchA (*hchA*), hypothetical proteins (*hypoth*), lactose operon repressor (*lacI*), hydroxyparomomycin C oxidase (*livQ*), propane monoxygenase large subunit (*prmA*), reductase component (*prmb*), small subunit (*prmc*) and effector component (*prmd*), ECF RNA polymerase sigma factor SigH (*sigH*), E-2-(N-methylformamido)methylene)succinate hydrolase (*yihR*) and E-2-(N-methylformamido)methylene)succinate hydrolase (*tnnD*).

et al., 2005). The trend discovered in the current work suggests that bioaugmentation with cultures containing *Rhodococcus* would be more appropriate at sites dominated by TCE. In contrast, bioaugmentation with cultures containing *Mycobacterium* would be more appropriate at sites with significant cDCE contamination. At sites with mixed contamination (as is often the case), either would likely be effective, as both *Mycobacterium* and *Rhodococcus* enriched cultures degraded both chemicals. The ability to degrade multiple contaminants has also been observed by others. Specifically, ethane-utilizing *Mycobacterium* spp. strains TA5 and TA27 degraded a range of chlorinated chemicals, including TCE and cDCE (Hashimoto et al., 2000). Similarly, *Rhodococcus* strains (isolate *Rhodococcus* Sm-1 and *Rhodococcus rhodochrous* ATCC 21197) degraded a number of chlorinated chemicals, including TCE and cDCE (Malachowsky et al., 1994).

An interesting question concerns why did *Rhodococcus* and

Mycobacterium/Mycolicibacterium dominate in TCE and cDCE degrading cultures, respectively. A speculative hypothesis involves the presence of a catalase gene (*katA*) close to the propane monoxygenase operon in all five *Rhodococcus* MAGs (this gene was absent in all fifteen *Mycobacterium* MAGs). Catalases represent oxidative stress-scavenging systems that detoxify reactive oxygen species (such as hydroxyl radicals) in bacteria growing under aerobic conditions (Alzaidi et al., 2025). If TCE biodegradation stimulated hydroxyl radical formation, as has been noted for exposure to antibiotics (Kohanski et al., 2007), *Rhodococcus* would be better prepared to limit the toxic effects compared to *Mycobacterium/Mycolicibacterium*. Conversely, *Mycobacterium* may have dominated cDCE amended propanotrophic cultures due to their impressive diversity, as illustrated by the detection of multiple *Mycobacterium* MAGs in many samples and multiple trends (eight) for the propane monoxygenase flanking genes. Clearly, research is needed to

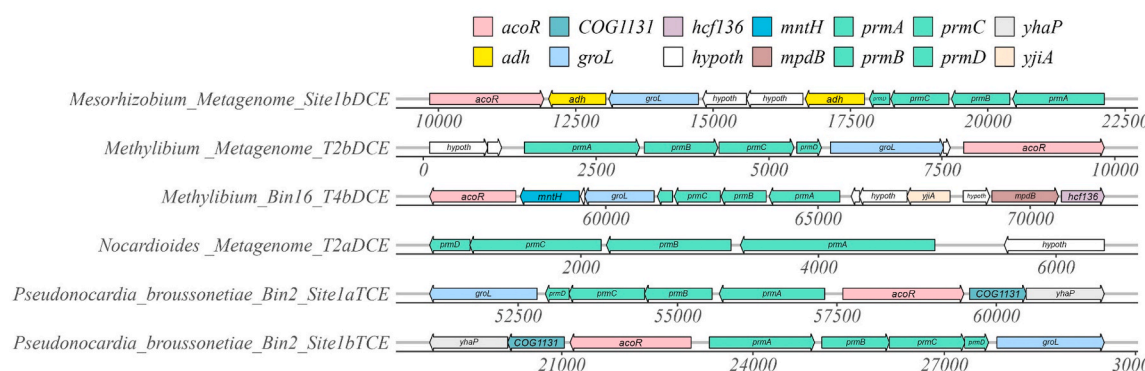


Fig. 15. Gene arrangement and length of the propane monoxygenase operon and surrounding genes from metagenomes (most similar to those from the genera *Mesorhizobium*, *Methylibium* and *Nocardioidees*) as well as metagenome assembled genomes classifying as *Methylibium* and *Pseudonocardia broussonetiae*. Four and two are from cDCE and TCE degrading microcosms, respectively. The encoded proteins are acetoin dehydrogenase operon transcriptional activator AcoR (*acoR*), alcohol dehydrogenase (*adh*), multidrug efflux system ATP-binding protein (*COG1131*), 60 kDa chaperonin (*groL*), Ycf48-like protein (*hcf136*), hypothetical proteins (*hypoth*), divalent metal cation transporter MntH (*mntH*), 2-methyl-1,2-propanediol dehydrogenase (*mpdB*), propane monoxygenase large subunit (*prmA*), reductase component (*prmB*), small subunit (*prnC*) and effector component (*prmD*), P-loop guanosine triphosphatase YjiA (*yjiA*) and putative protein YhaP (*yhaP*).

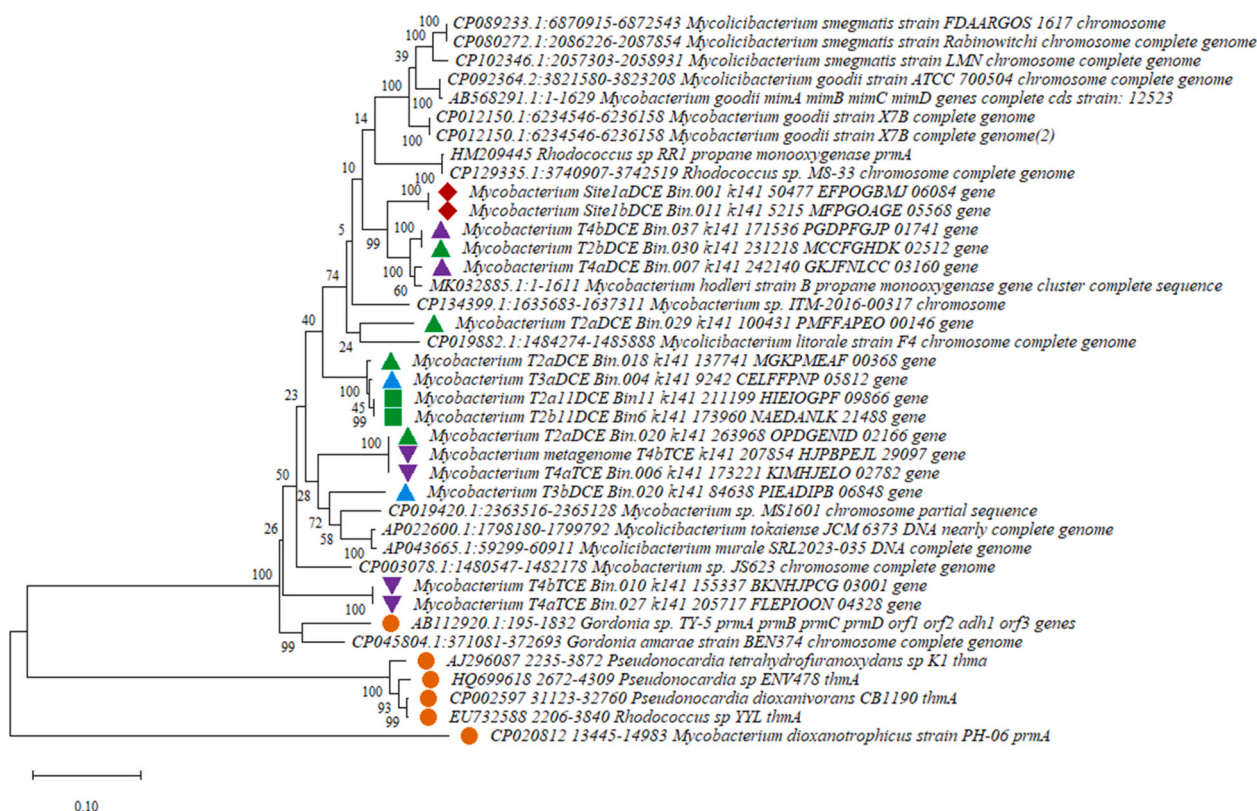


Fig. 16. Alignment of the propane monoxygenase large subunit (*prmA*) sequences from *Mycobacterium* MAGs and metagenomes along with the closest matches in GenBank and previously identified *prmA* and *thmA* sequences (orange symbols). The green, blue, purple and red symbols represent sequences from soils T2, T3, T4 and Site 1 respectively. The upward and downward triangles represent cDCE and TCE amended samples, respectively. Two MAGs were from 1,1-DCE degrading microcosms (green squares). The phylogeny was inferred using the Maximum Likelihood method and Jukes-Cantor (1969) model of nucleotide substitutions and the tree with the highest log likelihood (−15,639.88) is shown. The percentage of replicate trees in which the associated taxa clustered together (1000 replicates) is shown next to the branches. The initial tree for the heuristic search was selected by choosing the tree with the superior log-likelihood between a Neighbor-Joining (NJ) tree and a Maximum Parsimony (MP) tree. The NJ tree was generated using a matrix of pairwise distances computed using the Jukes-Cantor (1969) model. The MP tree had the shortest length among 10 MP tree searches, each performed with a randomly generated starting tree. The analytical procedure encompassed 39 coding nucleotide sequences using 1st, 2nd, 3rd, and non-coding positions with 1695 positions in the final dataset. Evolutionary analyses were conducted in MEGA12 utilizing up to 4 parallel computing threads.

investigate these hypotheses.

The biodegradation trends for 1,1-DCE in the current study were clearly different to those for cDCE and TCE. Limitations associated with 1,1-DCE biodegradation have also been noted by others. 1,1-DCE

biodegradation has previously been linked to toxicity associated with 1,1-DCE transformation products (Dolan and McCarty, 1995). For example, a study with two isoprene utilizing bacteria, *Alcaligenes denitrificans* ssp. *xylosoxidans* JE 75 and *Rhodococcus erythropolis* JE 77,

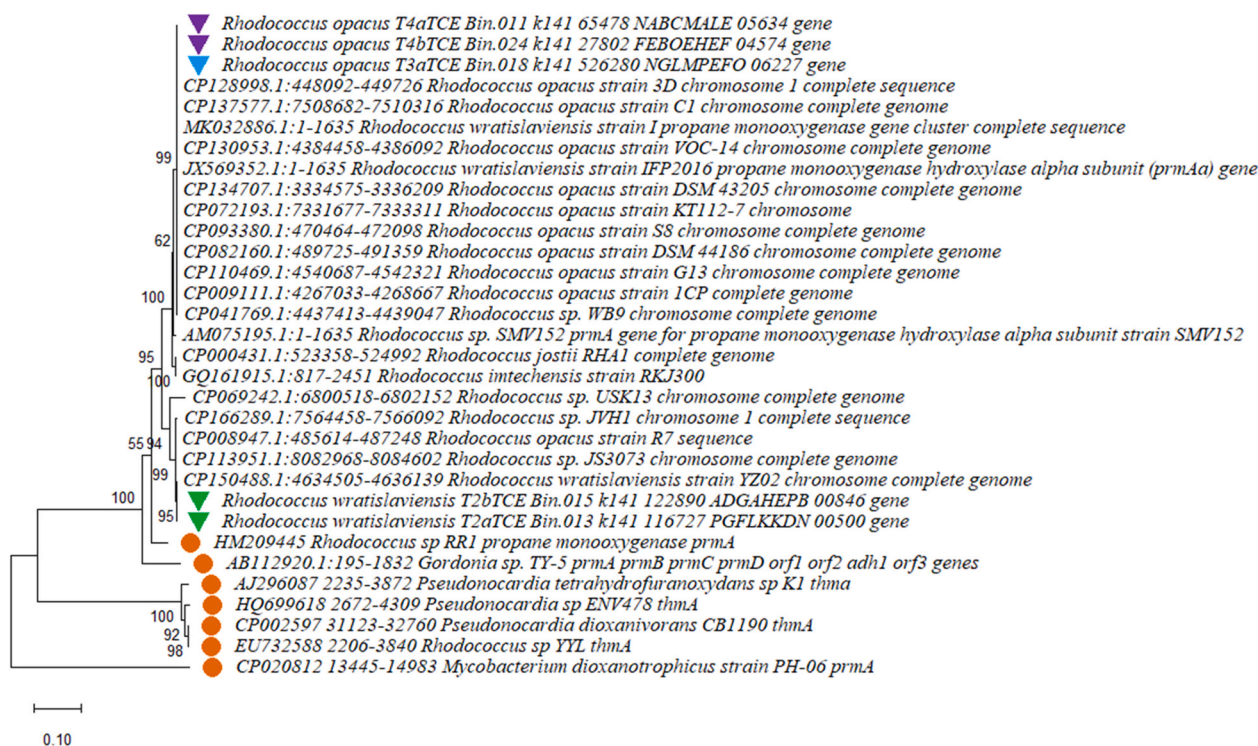


Fig. 17. Alignment of the propane monooxygenase large subunit (*prmA*) sequences from *Rhodococcus* MAGs along with the closest matches in GenBank and previously identified *prmA* and *thmA* sequences (orange symbols). The green, blue and purple symbols represent sequences from soils T2, T3 and T4 and respectively. All MAGs were from the TCE amended samples. The phylogeny was inferred using the Maximum Likelihood method and Jukes-Cantor (1969) model of nucleotide substitutions and the tree with the highest log likelihood (−8784.18) is shown. The percentage of replicate trees in which the associated taxa clustered together (1000 replicates) is shown next to the branches. The initial tree for the heuristic search was selected by choosing the tree with the superior log-likelihood between a Neighbor-Joining (NJ) tree and a Maximum Parsimony (MP) tree. The NJ tree was generated using a matrix of pairwise distances computed using the Jukes-Cantor (1969) model. The MP tree had the shortest length among 10 MP tree searches, each performed with a randomly generated starting tree. The analytical procedure encompassed 32 coding nucleotide sequences using 1st, 2nd, 3rd, and non-coding positions with 1696 positions in the final dataset. Evolutionary analyses were conducted in MEGA12 utilizing up to 4 parallel computing threads.

found greater cytotoxic effects from 1,1-DCE compared to other chlorinated ethenes (Ewers et al., 1990). In a methanotrophic mixed culture, cell inactivation from 1,1-DCE transformation was greater than for any of the other chloroethenes tested (tDCE, cDCE, TCE, VC) (Dolan and McCarty, 1995). In studies with the isolate *Comamonas testosteroni* RF2 (using phenol and sodium lactate as the primary substrates), only a low concentration of 1,1-DCE (0.25 mg/L) was degraded completely (Zalesak et al., 2017). Partial removal occurred at 0.37 mg/L 1,1-DCE and negligible removal was observed at 1.77 mg/L 1,1-DCE (Zalesak et al., 2017). In a butane grown mixed culture, 1,1-DCE transformation caused greater cell inactivation than the transformation of other chlorinated ethenes (Kim et al., 2000). Similarly, microcosm experiments and modeling studies with a butane grown culture (primarily composed of a *Rhodococcus* sp.) showed high 1,1-DCE transformation product toxicity (Semprini et al., 2007). Further, the authors indicated the loss of performance in the bioaugmentation field test may also have been associated with 1,1-DCE transformation toxicity (Semprini et al., 2007). The cometabolic transformation of a mixture of 1,1,1-TCA and 1,4-dioxane was also negatively impacted when 1,1-DCE (100–250 µg/L) was added to laboratory columns filled with beads containing *Rhodococcus rhodochrous* 21,198 (Azizian and Semprini, 2022). Toxicity from the transformation of 1,1-DCE was also thought to be responsible for a stall in the cometabolic transformation of cDCE, 1,4-dioxane and 1,1,1-TCA in groundwater microcosm studies (Rolston et al., 2022). 1,1-DCE transformation toxicity was also associated with the limited decline in cDCE, 1,1-DCA and 1,4-dioxane at a field site, undergoing methane and isobutane injections, as well as bioaugmentation (isobutane oxidizer *Rhodococcus aetherivorans* ENV493 and mixed methanotrophic consortium ENV494M) (Danko et al., 2023).

In contrast, *Azoarcus* sp. DD4 sustained the concurrent co-oxidation of 1,4-dioxane and 1,1-DCE using propane as the primary substrate (Deng et al., 2018b). Further, two psychrophilic consortia degraded 1,4-dioxane, 1,1-DCE and numerous other chlorinated aliphatic hydrocarbons when amended with propane (Antunes and Li, 2024).

The toxicity associated with 1,1-DCE has important implications for the remediation of mixed contaminant sites. Specifically, it may be necessary to have multiple rounds of bioaugmentation during the remediation project to ensure acceptable levels of active propanotrophs. Following the removal of 1,1-DCE, repeated rounds of bioaugmentation may no longer be needed. Also, it may be more effective to bioaugment with different cultures, first with efficient 1,1-DCE degraders, followed by those more capable of degrading the remaining contaminants (e.g. *Rhodococcus* for TCE). In sum, any bioaugmentation project at a 1,1-DCE contaminated site would benefit from additional information on which propanotrophs are more resistant to the toxic effects of 1,1-DCE.

Here, 1,1-DCE biodegradation was only noted in two of the four cDCE degrading cultures and in one of the four TCE degrading cultures. A trend perhaps indicating fewer microorganisms are capable of degrading 1,1-DCE compared to cDCE and TCE. When propane was not amended to three replicates, 1,1-DCE degradation did not occur, suggesting removal was associated with propanotrophs. Both 1,1-DCE degrading cultures were dominated by *Pseudonocardia*, however, all *Pseudonocardia* MAGs contained a truncated (thus, non-functional) propane monooxygenase large subunit. Different hypotheses may explain this finding. One hypothesis being that 1,1-DCE and propane biodegradation may be facilitated by oxygenases in *Pseudonocardia* other than the classic group 5 or 6 propane monooxygenases. For example, all four *Pseudonocardia* MAGs in the 1,1-DCE degrading



Fig. 18. Alignment of the propane monoxygenase large subunit (*prmA*) sequences from *Pseudonocardia broussonetiae* and *Methylilium* MAGs and metagenomes (similar to *Mesorhizobium*, *Nocardioides* and *Methylilium* *prmA* sequences) along with the closest matches in GenBank and previously identified *prmA* and *thmA* sequences (orange symbols). The green, purple and red symbols represent sequences from soils T2, T4 (triangles, DCE amended only) and Site 1 (diamonds, TCE and DCE amended) respectively. The phylogeny was inferred using the Maximum Likelihood method and Jukes-Cantor (1969) model of nucleotide substitutions and the tree with the highest log likelihood (−15,066.73) is shown. The percentage of replicate trees in which the associated taxa clustered together (100 replicates) is shown next to the branches. The initial tree for the heuristic search was selected by choosing the tree with the superior log-likelihood between a Neighbor-Joining (NJ) tree and a Maximum Parsimony (MP) tree. The NJ tree was generated using a matrix of pairwise distances computed using the Jukes-Cantor (1969) model. The MP tree had the shortest length among 10 MP tree searches, each performed with a randomly generated starting tree. The analytical procedure encompassed 32 coding nucleotide sequences using 1st, 2nd, 3rd, and non-coding positions with 1707 positions in the final dataset. Evolutionary analyses were conducted in MEGA12 utilizing up to 4 parallel computing threads.

cultures contain multiple cytochromes and multiple putative cytochromes. Another hypothesis being *Mycobacterium* was responsible for 1,1-DCE degradation (two propane monoxygenase containing *Mycobacterium* MAGs were recovered) and *Pseudonocardia*, although unable to degrade propane, was dominant due to the consumption of propane or 1,1-DCE degradation products from *Mycobacterium*. To resolve this puzzle, future research will determine the upregulated genes during propane and 1,1-DCE degradation.

Notably, both propane and 1,1-DCE biodegradation trends slowed over multiple transfers. These results are consistent with the literature discussed above, concerning the toxic impacts of 1,1-DCE biodegradation, reducing the active degraders. However, following the depletion of 1,1-DCE, the majority of replicates (of both cultures) were capable of both propane and TCE biodegradation, indicating the toxic impacts were reversible.

Whole genome sequencing analysis indicated the presence of both group 5 and putative group 6 propane monoxygenase operons in the propanotrophic metagenomes and MAGs. The order of group 5 propane monoxygenase subunits (*prmA*BCD) in the current study, large subunit (*prmA*), reductase subunit (*prmB*), small subunit (*prmC*) and coupling protein (*prmD*), was the same as those previously reported in *Gordonia* sp. TY-5 (Kotani et al., 2003), *Rhodococcus* sp. strain RHA1 (Sharp et al.,

2007), *Rhodococcus* sp. strain BCP1 (Cappelletti et al., 2015), *Mycobacterium goodii* strain 12,523, *Mycobacterium smegmatis* mc² 155 (designated *mimABCD* in both) (Furuya et al., 2011), *Azoarcus* DD4 (Deng et al., 2019) and *Pseudonocardia* sp. TY-7 (Kotani et al., 2006). In the current study, MAGs containing the full operon of group 5 propane monoxygenase classified within the order *Actinomycetales* (*R. wratislaviensis*, *R. opacus*, *P. broussonetiae*, *Mycobacterium*), with one within the order *Burkholderiales* (*Methylilium*).

The current work also identified putative group 6 propane monoxygenase gene clusters and, consistent with previous research (on *Mycobacterium* and *Rhodococcus*), these contained a different subunit order compared to group 5 propane monoxygenase operons. Specifically, *Mycobacterium* sp. TY-6 contains the following subunit order for a group 6 propane monoxygenase: hydroxylase large subunit (*prmA*), hydroxylase small subunit (*prmC*), coupling protein (*prmD*), and reductase (*prmB*) (Kotani et al., 2006). *M. chubuense* strain NBB4 and *Mycobacterium* sp. strain ENV421 contain a similar gene order (Coleman et al., 2011; Masuda et al., 2012). *M. dioxanotrophicus* PH-06 also contains a group 6 propane monoxygenase in the following order: hydroxylase α subunit (*prmA*), hydroxylase β subunit (*prmB*), coupling protein (*prmC*), reductase (*prmD*) (Deng et al., 2018a) (although the names of the subunits are different, the order and enzymes they encode

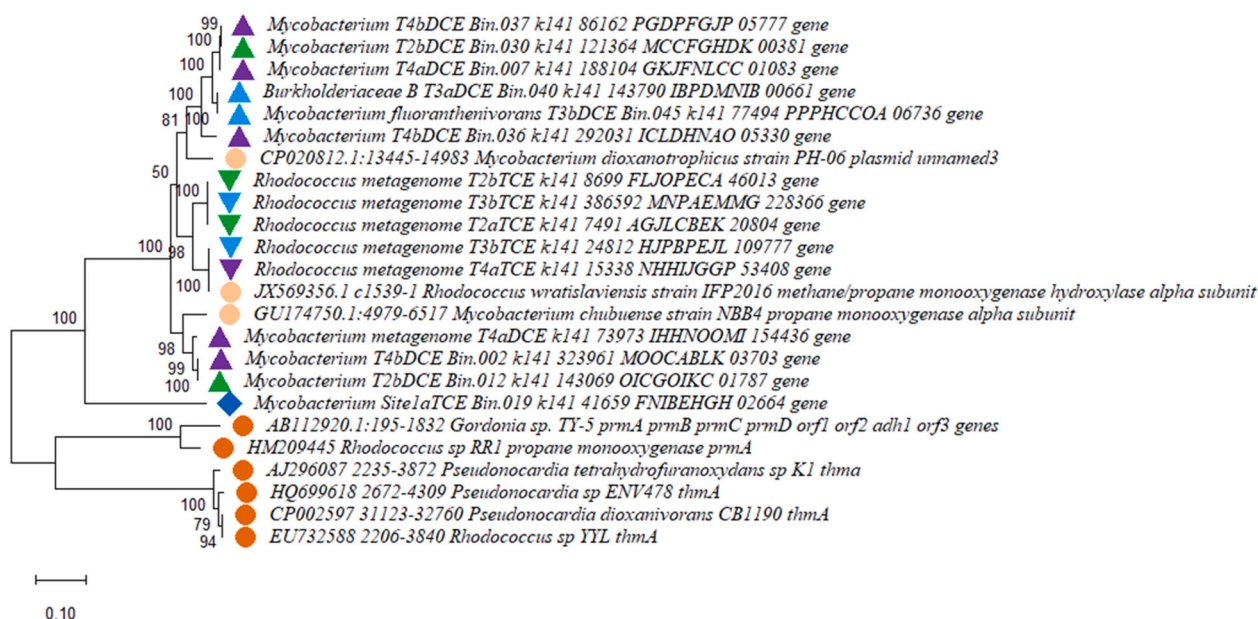


Fig. 19. Alignment of the propane monoxygenase large subunit (*prmA*) sequences from MAGs and metagenomes containing putative group 6 propane monoxygenases along with previously identified group five *prmA* and *thmA* sequences (orange symbols) and group six *prmA* sequences (light orange symbols). The green, blue, purple and dark blue symbols represent sequences from soils T2, T3, T4 and Site 1 respectively. The upward and downward triangles represent cDCE and TCE amended samples for the soils and the diamond symbol for a sample from Site 1 (TCE amended). The phylogeny was inferred using the Maximum Likelihood method and Jukes-Cantor (1969) model of nucleotide substitutions and the tree with the highest log likelihood (−11,762.66) is shown. The percentage of replicate trees in which the associated taxa clustered together (1000 replicates) is shown next to the branches. The initial tree for the heuristic search was selected by choosing the tree with the superior log-likelihood between a Neighbor-Joining (NJ) tree and a Maximum Parsimony (MP) tree. The NJ tree was generated using a matrix of pairwise distances computed using the Jukes-Cantor (1969) model. The MP tree had the shortest length among 10 MP tree searches, each performed with a randomly generated starting tree. The analytical procedure encompassed 24 coding nucleotide sequences using 1st, 2nd, 3rd, and non-coding positions with 1692 positions in the final dataset. Evolutionary analyses were conducted in MEGA12 utilizing up to 4 parallel computing threads.

are the same). The propane monoxygenase for *M. dioxanotrophicus* PH-06 is located on Plasmid_3 and is similar to gene clusters in *Rhodococcus wratislaviensis* IFP2016 and *M. chubuense* NBB4 (Auffret et al., 2015; He et al., 2017b). The alpha subunit of *M. dioxanotrophicus* PH-06 was similar to that from *Rhodococcus aetherivorans* BCP1 (Ferrari et al., 2024). The same order (and naming) was reported for a group 6 propane monoxygenase gene cluster in *M. vaccae* JOB5 (propane monoxygenase oxygenase α -subunit, *prmA*, oxygenase β -subunit, *prmB*, coupling protein *prmC*, reductase, *prmD*) (Chen et al., 2023). Notably, a recent study suggested *M. dioxanotrophicus* PH-06 is the donor bacterial species of the group 6 SDIMOs in the *Rhodococcus* genus (Ferrari et al., 2024). Consistent with the trends above, in the current study, putative group 6 propane monoxygenases were found in *Mycobacterium* MAGs (in cDCE degrading cultures and in one TCE degrading culture). In the metagenomes, group 6 alpha subunits most closely aligned with *Rhodococcus* (primarily in the TCE degrading cultures).

An important question concerns the relative importance of group 5 or group 6 in propane/cDCE/TCE/1,1-DCE biodegradation. Some insight can perhaps be obtained from a study that compared expression levels for one group 5 and one group 6 SDIMO in *Rhodococcus* sp. strain BCP1 following exposure to propane (Cappelletti et al., 2015). The expression of the large subunit for each was 1500 fold (group 5) and <85 fold (group 6) higher when grown on propane compared to succinate, indicating enhanced expression (and perhaps greater importance) of the group 5 SDIMO (Cappelletti et al., 2015). Clearly, additional research is needed to determine the role of group 5 versus group 6 propane monoxygenases in both propane and contaminant biodegradation.

Propane monoxygenase flanking genes from the *Rhodococcus opacus* MAGs in the current study are similar to those previously reported for *Rhodococcus* sp. strain BCP1, *Rhodococcus* sp. strain RHA1, and *R. opacus* PD630. Specifically, in those strains, downstream of *prmA* are two genes (a metal-dependent hydrolase, a metal-sulfur cluster biosynthetic

enzyme) followed by an alcohol dehydrogenase, and a GroEL chaperon (Cappelletti et al., 2015). In the current study, the two genes in parenthesis were hypothetical genes. Upstream genes, in both current and previous work, involved those coding for the regulation of transcription and a catalase (Cappelletti et al., 2015). The *R. wratislaviensis* MAGs gene clusters in the current work included another alcohol dehydrogenase downstream of *prmA* as well as another transcriptional regulator. As indicated in the results section, the flanking genes in the *Mycobacterium* MAGs illustrated more diversity compared to those from the *Rhodococcus* MAGs. Although, in general, the majority contained one or more alcohol dehydrogenases, a gene coding a chaperonin protein and transcriptional regulators. The flanking genes from the other MAGs in the current study also contained similar genes. The majority of the group 6 propane monoxygenase flanking genes in the metagenomes illustrated the same three downstream genes (coding an aldehyde dehydrogenase, an alcohol dehydrogenase and a chaperonin protein). Upstream, many also contained a transcriptional regulator (*narL*, *desR*, *vraR*).

Possible roles of such flanking genes have previously been reported. For example, downstream alcohol dehydrogenases may be involved in 2-propanol oxidation, as has been reported for the propanotroph *Gordonia* sp. strain TY-5 (transcribed in response to propane and 2-propanol) (Kotani et al., 2006). The groEL-encoded chaperonin protein has previously been associated with the productive folding of SDIMO hydroxylase subunits (Furuya et al. 2011, 2013). This gene has previously been associated with both group 5 (*Gordonia* sp. strain TY-5, *Rhodococcus* sp. strain BCP1, *Rhodococcus* sp. strain RHA1, *R. opacus* PD630) and group 6 (*M. vaccae* JOB5, *M. dioxanotrophicus* PH-06, *M. chubuense* NBB4, *Rhodococcus* sp. strain BCP1) propane monoxygenase gene clusters (Cappelletti et al., 2015; Chen et al., 2023; Deng et al., 2018a; Kotani et al., 2006). Catalase (*katA*) has been associated with protecting microorganisms from reactive oxygen species (Albandari et al., 2025). The transcription regulators are likely important for the expression of these

gene clusters. Future research should be directed towards determining which genes are upregulated during propane and chlorinated ethene biodegradation.

5. Conclusions

In summary, a key finding in the current study is the importance of genera *Mycobacterium*/*Mycolicibacterium* and *Rhodococcus* for cDCE and TCE biodegradation, respectively. Such information could be used to tailor bioaugmentation efforts to individual sites, dependent on the dominant contaminants. Further, the data suggest a more limited number of propanotrophic phylotypes are capable of 1,1-DCE biodegradation. More information is needed on which propanotrophs are more resistant to the toxic effects of 1,1-DCE degradation. The molecular data indicated propane monooxygenases and flanking genes were more consistent between *Rhodococcus* MAGs compared to *Mycobacterium* MAGs. The mixed cultures contained both group 5 and putative group 6 propane monooxygenases and, thus, both may be involved in propane and chlorinated ethene degradation. Therefore, qPCR on contaminated groundwater and sediment should target both groups of genes. This work is the first to provide in depth metagenomic information on both gene clusters and flanking genes, contributing to a better understanding of the functional basis for propane oxidation. The mixed cultures developed here have potential for use in bioaugmentation efforts at contaminated sites due to their ability to degrade multiple chlorinated ethenes along with 1,4-dioxane.

CRediT authorship contribution statement

Mohsen Faghihinezhad: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Zohre Eshghdoostkhatami:** Conceptualization, Data curation, Investigation, Methodology, Validation, Visualization, Writing – review & editing. **Alison M. Cupples:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Supervision, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2026.129957>.

Data availability

Whole genome sequencing data has been submitted to NCBI.

References

- Adamson, D.T., Anderson, R.H., Mahendra, S., Newell, C.J., 2015. Evidence of 1,4-dioxane attenuation at groundwater sites contaminated with chlorinated solvents and 1,4-dioxane. *Environ. Sci. Technol.* 49 (11), 6510–6518.
- Adamson, D.T., Mahendra, S., Walker Jr., K.L., Rauch, S.R., Sengupta, S., Newell, C.J., 2014. A multisite survey to identify the scale of the 1,4-dioxane problem at contaminated groundwater sites. *Environ. Sci. Technol. Lett.* 1 (5), 254–258.
- Albandari, A., Lajoie, M., Kwon, D.H., 2025. Differential roles of catalase-encoding genes (*katA* and *katB*) in response to oxidative stress and antibiotics in *Pseudomonas aeruginosa*. *Gene Reports*, 102226.
- Alexander, M., 1994. *Biodegradation and Bioremediation*. Academic Press Inc., San Diego, California, p. 303.
- Alneberg, J., Bjarnason, B.S., de Bruijn, I., Schirmer, M., Quick, J., Ijaz, U.Z., Lahti, L., Loman, N.J., Andersson, A.F., Quince, C., 2014. Binning metagenomic contigs by coverage and composition. *Nat. Methods* 11 (11), 1144–1146.
- Altschul, S.F., Gish, W., Miller, W., Myers, E.W., Lipman, D.J., 1990. Basic local alignment search tool. *J. Mol. Biol.* 215 (3), 403–410.
- Alzaidi, A., Lajoie, M., Kwon, D.H., 2025. Differential roles of catalase-encoding genes (*katA* and *katB*) in response to oxidative stress and antibiotics in *Pseudomonas aeruginosa*. *Gene Reports* 40, 102226.
- Anderson, R.H., Anderson, J.K., Bower, P.A., 2012. Co-occurrence of 1,4-dioxane with trichloroethylene in chlorinated solvent groundwater plumes at US Air Force installations: fact or fiction. *Integrated Environ. Assess. Manag.* 8 (4), 731–737.
- Andrews, S., 2010. FastQC: a quality control tool for high throughput sequence data. <https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>.
- Antunes, J.M.D., Li, M., 2024. Cometabolic biodegradation of 1, 4-dioxane and co-occurring chlorinated aliphatic hydrocarbons by psychrophilic propanotrophs enriched with a new cluster of group-6 soluble di-iron monooxygenases. *Environ. Sci. Technol. Lett.*
- Arkin, A.P., Cottingham, R.W., Henry, C.S., Harris, N.L., Stevens, R.L., Maslov, S., Dehal, P., Ware, D., Perez, F., Canon, S., Sneddon, M.W., Henderson, M.L., Riehl, W. J., Murphy-Olson, D., Chan, S.Y., Kamimura, R.T., Kumari, S., Drake, M.M., Brettin, T.S., Glass, E.M., Chivian, D., Gunter, D., Weston, D.J., Allen, B.H., Baumohl, J., Best, A.A., Bowen, B., Brenner, S.E., Bun, C.C., Chandonia, J.-M., Chiu, J.-M., Colasanti, R., Conrad, N., Davis, J.J., Davison, B.H., DeJongh, M., Devoid, S., Dietrich, E., Dubchak, I., Edirisinghe, J.N., Fang, G., Faria, J.P., Frybarger, P.M., Gerlach, W., Gerstein, M., Greiner, A., Gurtowski, J., Haun, H.L., He, F., Jain, R., Joachimiak, M.P., Keegan, K.P., Kondo, S., Kumar, V., Land, M.L., Meyer, F., Mills, M., Novichkov, P.S., Oh, T., Olsen, G.J., Olson, R., Parrello, B., Pasternak, S., Pearson, E., Poon, S.S., Price, G.A., Ramakrishnan, S., Ranjan, P., Ronald, P.C., Schatz, M.C., Seaver, S.M.D., Shukla, M., Sutormin, R.A., Syed, M.H., Thomason, J., Tintle, N.L., Wang, D., Xia, F., Yoo, H., Yoo, S., Yu, D., 2018. KBase: the United States Department of energy systems biology knowledgebase. *Nat. Biotechnol.* 36 (7), 566–569.
- Auffret, M.D., Yergeau, E., Labbe, D., Fayolle-Guichard, F., Greer, C.W., 2015. Importance of *Rhodococcus* strains in a bacterial consortium degrading a mixture of hydrocarbons, gasoline, and diesel oil additives revealed by metatranscriptomic analysis. *Appl. Microbiol. Biotechnol.* 99 (5), 2419–2430.
- Aziz, R.K., Bartels, D., Best, A.A., DeJongh, M., Disz, T., Edwards, R.A., Formsma, K., Gerdes, S., Glass, E.M., Kubal, M., Meyer, F., Olsen, G.J., Olson, R., Osterman, A.L., Overbeek, R.A., McNeil, L.K., Paarmann, D., Paczian, T., Parrello, B., Pusch, G.D., Reich, C., Stevens, R., Vassieva, O., Vonstein, V., Wilke, A., Zagnitko, O., 2008. The RAST server: rapid annotations using subsystems technology. *BMC Genom.* 9 (1), 75.
- Azizian, M.F., Semprini, L., 2022. Aerobic cometabolism of chlorinated solvents and 1,4-dioxane in continuous-flow columns packed with gellan-gum hydrogels coencapsulated with ATCC Strain 21198 and TBOS or T2BOS as slow-release compounds. *ACS ES&T Eng.* 2, 1531–1543.
- Baskaran, D., Paul, T., Kannan, P., Krithivasan, M., Devanesan, M.G., Rajamanickam, R., 2020. Batch degradation of trichloroethylene using oleaginous *Rhodococcus opacus* in a two-phase partitioning bioreactor and kinetic study. *Bioresour. Technol. Rep.* 11. <https://doi.org/10.1016/j.biteb.2020.100437>.
- Bealessio, A.D., Chen, W., Krippaehne, K.J., Murnane, R.A., Hyman, M.R., Semprini, L., 2023. Alcohol-dependent cometabolic degradation of chlorinated aliphatic hydrocarbons and 1,4-dioxane by *Rhodococcus rhodochrous* strain ATCC 21198. *Environ. Eng. Sci.* 40 (11), 494–505.
- Bell, C.H., Wong, J., Parsons, K., Semel, W., McDonough, J., Gerbe, K., 2022. First full-scale in situ propane biosparging for co-metabolic bioremediation of 1,4-dioxane. *Ground Water Monit. Remed.* 42, 54–66.
- Bolger, A.M., Lohse, M., Usadel, B., 2014. Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics* 30 (15), 2114–2120.
- Bouwer, E.J., Rittmann, B.E., McCarty, P.L., 1981. Anaerobic degradation of halogenated 1- and 2-carbon organic compounds. *Environ. Sci. Technol.* 15 (5), 596–599.
- Bowers, R.M., Kyrpides, N.C., Stepanauskas, R., Harmon-Smith, M., Doud, D., Reddy, T. B.K., Schulz, F., Jarett, J., Rivers, A.R., Eloie-Padros, E.A., Tringe, S.G., Ivanova, N. N., Copeland, A., Clum, A., Becraft, E.D., Malmstrom, R.R., Birren, B., Podar, M., Bork, P., Weinstock, G.M., Garrity, G.M., Dodsworth, J.A., Yooshep, S., Sutton, G., Glöckner, F.O., Gilbert, J.A., Nelson, W.C., Hallam, S.J., Jungbluth, S.P., Ettema, T.J. G., Tighe, S., Konstantinidis, K.T., Liu, W.-T., Baker, B.J., Rattei, T., Eisen, J.A., Hedlund, B., McMahon, K.D., Fierer, N., Knight, R., Finn, R., Cochrane, G., Karsch-Mizrachi, I., Tyson, G.W., Rinke, C., Kyrpides, N.C., Schriml, L., Garrity, G.M., Hugenholtz, P., Sutton, G., Yilmaz, P., Meyer, F., Glöckner, F.O., Gilbert, J.A., Knight, R., Finn, R., Cochrane, G., Karsch-Mizrachi, I., Lapidus, A., Meyer, F., Yilmaz, P., Parks, D.H., Murat Eren, A., Schriml, L., Banfield, J.F., Hugenholtz, P., Woyke, T., The Genome Standards, C., 2017. Minimum information about a single

- amplified genome (MISAG) and a metagenome-assembled genome (MIMAG) of bacteria and archaea. *Nat. Biotechnol.* 35 (8), 725–731.
- Cansdale, A., Chong, J.P.J., 2024. MAGqual: a stand-alone pipeline to assess the quality of metagenome-assembled genomes. *Microbiome* 12 (1), 226.
- Cappelletti, M., Pinelli, D., Fedi, S., Zannoni, D., Frascari, D., 2018. Aerobic co-metabolism of 1,1,2,2-tetrachloroethane by *Rhodococcus aetherivorans* TPA grown on propane: kinetic study and bioreactor configuration analysis. *J. Chem. Technol. Biotechnol.* 93, 155–165.
- Cappelletti, M., Presentato, A., Milazzo, G., Turner, R.J., Fedi, S., Frascari, D., Zannoni, D., 2015. Growth of *Rhodococcus* sp. strain BCP1 on gaseous n-alkanes: new metabolic insights and transcriptional analysis of two soluble di-iron monooxygenase genes. *Front. Microbiol.* 6, 393.
- Chaumeil, P.A., Mussig, A.J., Hugenholtz, P., Parks, D.H., 2019. GTDB-Tk: a toolkit to classify genomes with the Genome Taxonomy Database. *Bioinformatics* 36 (6), 1925–1927.
- Chen, Y., Ren, H., Kong, X., Wu, H., Lu, Z., 2023. A multicomponent propane monooxygenase catalyzes the initial degradation of methyl tert-butyl ether in *Mycobacterium vaccae* JOB5. *Appl. Environ. Microbiol.* 89 (10), e0118723.
- Chivian, D., Jungbluth, S.P., Dehal, P.S., Wood-Charlson, E.M., Canon, R.S., Allen, B.H., Clark, M.M., Gu, T., Land, M.L., Price, G.A., Riehl, W.J., Sneddon, M.W., Sutormin, R., Zhang, Q., Cottingham, R.W., Henry, C.S., Arkin, A.P., 2023. Metagenome-assembled genome extraction and analysis from microbiomes using KBase. *Nat. Protoc.* 18 (1), 208–238.
- Coleman, N.V., Yau, S., Wilson, N.L., Nolan, L.M., Migocki, M.D., Ly, M.A., Crossett, B., Holmes, A.J., 2011. Untangling the multiple monooxygenases of *Mycobacterium chubuense* strain NBB4, a versatile hydrocarbon degrader. *Environ Microbiol Rep* 3 (3), 297–307.
- Cupples, A.M., 2024. Propane monooxygenases in soil associated metagenomes align most closely to those in the genera *Kribbella*, *Amycolatopsis*, *Bradyrhizobium*, *Paraburkholderia* and *Burkholderia*. *Curr. Microbiol.* 81 (10), 314.
- Dabrock, B., Riedel, J., Bertram, J., Gottschalk, G., 1992. Isopropylbenzene (cumene)—a new substrate for the isolation of trichloroethene-degrading bacteria. *Arch. Microbiol.* 158 (1), 9–13.
- Danko, A., Lippincott, D., Hatzinger, P., Lavorgna, G., Semprini, L., Hyman, M.R., 2023. Evaluation of a Novel Multiple Primary Substrate (MPS) Cometabolic Approach for in Situ Bioremediation of 1,4-dioxane and Chlorinated Solvents in Groundwater. ESTCP Project ER-201733. Final Report.
- Deng, D., Li, F., Ye, L., Li, M., 2019. Complete genome sequence of *Azoarcus* sp. Strain DD4, a gram-negative propanotroph that degrades 1,4-dioxane and 1,1-dichloroethylene. *Microbiol. Resour. Announc.* 8 (33).
- Deng, D.Y., Li, F., Li, M.Y., 2018a. A novel propane monooxygenase initiating degradation of 1,4-dioxane by *Mycobacterium dioxanotrophicus* PH-06. *Environmental Science and Technology Letters* 5 (2), 86–91.
- Deng, D.Y., Li, F., Wu, C., Li, M.Y., 2018b. Synchronic biotransformation of 1,4-dioxane and 1,1-dichloroethylene by a Gram-negative propanotroph *Azoarcus* sp DD4. *Environ. Sci. Technol. Lett.* 5 (8), 526–+.
- Divine, C., Bell, C.H., Heintz, M.B., Lorenz, A., Vallin, P., Favero, D., Gerber, K., Zheng, C., Rittmann, B., 2024. Advances in remediation solutions: new developments and opportunities in 1,4-dioxane biological treatment. *Groundwater Monitoring & Remediation*. <https://doi.org/10.1111/gwmr.12649>.
- Doherty, R.E., 2000. A history of the production and use of carbon tetrachloride, tetrachloroethylene, trichloroethylene and 1,1,1-trichloroethane in the United States: part 1. Historical background; carbon tetrachloride and tetrachloroethylene. *J Environ Forensics* 1, 69–81.
- Dolan, M.E., McCarty, P.L., 1995. Methanotrophic chloroethene transformation capacities and 1,1-dichloroethene transformation product toxicity. *Environ. Sci. Technol.* 29 (11), 2741–2747.
- EPA, 2026. EPA On-line tools for site assessment calculation. <https://www3.epa.gov/ceampub/learn2model/part-two/onsite/esthenry.html>.
- Eshghdoostkhatami, Z., Li, Z., Faghihinezhad, M., Cupples, A.M., 2025. Characterization of propanotrophic enrichments from agricultural soils capable of 1,4-dioxane biodegradation to sub-mg/L levels. *Sci. Total Environ.* 1005, 180824.
- Ewers, J., Freier-Schrieder, D., Knackmuss, H.-J., 1990. Selection of trichloroethene (TCE) degrading bacteria that resist inactivation by TCE. *Arch. Microbiol.* 154, 410–413.
- Faghihinezhad, Mohsen, Eshghdoostkhatami, Zohre, Bernstein, Anat, Cupples, Alison M., 2025. Identification of the dominant methanotrophs in trichloroethene degrading enrichment cultures from multiple sources. *J. Hazardous Materials* 499, 140268.
- Fatthepure, B.Z., Elango, V.K., Singh, H., Bruner, M.A., 2005. Bioaugmentation potential of a vinyl chloride-assimilating *Mycobacterium* sp., isolated from a chloroethene-contaminated aquifer. *FEMS Microbiol. Lett.* 248 (2), 227–234.
- Ferrari, E., Di Benedetto, G., Firrincieli, A., Presentato, A., Frascari, D., Cappelletti, M., 2024. Unravelling the role of the group 6 soluble di-iron monooxygenase (SDIMO) SmoABCD in alkane metabolism and chlorinated alkane degradation. *Microb. Biotechnol.* 17 (5), e14453.
- Frydman, J., 2001. Folding of newly translated proteins *in vivo*: the role of molecular chaperones. *Annu. Rev. Biochem.* 70, 603–647.
- Furiya, T., Hayashi, M., Semba, H., Kino, K., 2013. The mycobacterial binuclear iron monooxygenases require a specific chaperonin-like protein for functional expression in a heterologous host. *FEBS J.* 280 (3), 817–826.
- Furiya, T., Hirose, S., Osanai, H., Semba, H., Kino, K., 2011. Identification of the monooxygenase gene clusters responsible for the regioselective oxidation of phenol to hydroquinone in mycobacteria. *Appl. Environ. Microbiol.* 77 (4), 1214–1220.
- Gossett, J.M., 1987. Measurement of Henry's law constants for C1 and C2 chlorinated hydrocarbons. *Environ. Sci. Technol.* 21 (2), 202–208.
- Hand, S., Wang, B., Chu, K.H., 2015. Biodegradation of 1,4-dioxane: effects of enzyme inducers and trichloroethylene. *Sci. Total Environ.* 520, 154–159.
- Hanson, J.R., Ackerman, C.E., Scow, K.M., 1999. Biodegradation of methyl tert-butyl ether by a bacterial pure culture. *Appl. Environ. Microbiol.* 65 (11), 4788–4792.
- Hartmans, S., De Bont, J.A., 1992. Aerobic vinyl chloride metabolism in *Mycobacterium aurum* L1. *Appl. Environ. Microbiol.* 58 (4), 1220–1226.
- Hashimoto, A., Iwasaki, K., Nakasugi, N., Nakajima, N., Yagi, O., 2000. Degradation of trichloroethylene and related compounds by *Mycobacterium* spp. isolated from soil. *Clean Prod. Process.* 2, 167–173.
- Hatzinger, P.B., 2022. Applying cometabolism for treatment of traditional and emerging contaminants at DoD sites. ESTCP and SERDP Webinar. <https://serdp-estcp.mil/webinars/details/bb647f30-d44e-473c-867b-2e64034f430a>.
- Hatzinger, P.B., Banerjee, R., Rezes, R., Streger, S.H., McClay, K., Schaefer, C.E., 2017. Potential for cometabolic biodegradation of 1,4-dioxane in aquifers with methane or ethane as primary substrates. *Biodegradation* 28, 453–468.
- Hatzinger, P.B., Streger, S.H., Begley, J.F., 2015. Enhancing aerobic biodegradation of 1,2-dibromoethane in groundwater using ethane or propane and inorganic nutrients. *J. Contam. Hydrol.* 172, 61–70.
- He, Y., Mathieu, J., Yang, Y., Yu, P., da Silva, M.L.B., Alvarez, P.J.J., 2017a. 1,4-Dioxane biodegradation by *Mycobacterium dioxanotrophicus* PH-06 is associated with a group-6 soluble di-iron monooxygenase. *Environmental Science and Technology Letters* 4 (11), 494–499.
- He, Y., Wei, K., Si, K., Mathieu, J., Li, M., Alvarez, P.J.J., 2017b. Whole-genome sequence of the 1,4-dioxane-degrading bacterium *Mycobacterium dioxanotrophicus* PH-06. *Genome Announc.* 5 (35).
- Hopkins, G.D., McCarty, P.L., 1995. Field evaluation of in situ aerobic cometabolism of trichloroethylene and three dichloroethylene isomers using phenol and toluene as the primary substrates. *Environ. Sci. Technol.* 29 (6), 1628–1637.
- Horst, J.F., Bell, C.H., Lorenz, A., Heintz, M., Miao, Y., Saling, J., Favero, D., Mahendra, S., 2019. Bioremediation of 1,4-dioxane: successful demonstration of in situ and ex situ approaches. *Groundwater Monitoring & Remediation* 39, 15–24.
- Horvath, R.S., 1972. Microbial co-metabolism and the degradation of organic compounds in nature. *Bacteriol. Rev.* 36 (2), 146.
- Kang, D.D., Froula, J., Egan, R., Wang, Z., 2015. MetaBAT, an efficient tool for accurately reconstructing single genomes from complex microbial communities. *PeerJ* 3, e1165.
- Karaoz, U., Brodie, E.L., 2022. microTrait: a toolset for a trait-based representation of microbial genomes. *Front Bioinform* 2, 918853.
- Karigar, C.S., Rao, S.S., 2011. Role of microbial enzymes in the bioremediation of pollutants: a review. *Enzym. Res.*, 2011.
- Kim, Y., Arp, D.J., Semprini, L., 2000. Chlorinated solvent cometabolism by butane-grown mixed culture. *Journal of Environmental Engineering ASCE* 126 (10), 934–942.
- Kohanski, M.A., Dwyer, D.J., Hayete, B., Lawrence, C.A., Collins, J.J., 2007. A common mechanism of cellular death induced by bactericidal antibiotics. *Cell* 130 (5), 797–810.
- Kotani, T., Kawashima, Y., Yurimoto, H., Kato, N., Sakai, Y., 2006. Gene structure and regulation of alkane monooxygenases in propane-utilizing *Mycobacterium* sp. TY-6 and *Pseudonocardia* sp. TY-7. *J. Biosci. Bioeng.* 102 (3), 184–192.
- Kotani, T., Yamamoto, T., Yurimoto, H., Sakai, Y., Kato, N., 2003. Propane monooxygenase and NAD⁺-dependent secondary alcohol dehydrogenase in propane metabolism by *Gordonia* sp. strain TY-5. *J. Bacteriol.* 185 (24), 7120–7128.
- Kumar, S., Stecher, G., Suleski, M., Sanderford, M., Sharma, S., Tamura, K., 2024. MEGA12: molecular evolutionary genetic analysis version 12 for adaptive and green computing. *Mol. Biol. Evol.* 41 (12).
- Langmead, B., Salzberg, S.L., 2012. Fast gapped-read alignment with Bowtie 2. *Nat. Methods* 9 (4), 357–359.
- Le, N.B., Coleman, N.V., 2011. Biodegradation of vinyl chloride, cis-dichloroethene and 1,2-dichloroethane in the alkane/alkane-oxidising *Mycobacterium* strain NBB4. *Biodegradation* 22 (6), 1095–1108.
- Li, D., Liu, C.M., Luo, R., Sadakane, K., Lam, T.W., 2015. MEGAHIT: an ultra-fast single-node solution for large and complex metagenomics assembly via succinct de Bruijn graph. *Bioinformatics* 31 (10), 1674–1676.
- Lippincott, D., Streger, S.H., Schaefer, C.E., Hinkle, J., Stormo, J., Steffan, R.J., 2015. Bioaugmentation and propane bioparging for in situ biodegradation of 1,4-dioxane. *Ground Water Monit. Remed.* 35 (2), 81–92.
- Malachowsky, K.J., Phelps, T.J., Teboli, A.B., Minnikin, D.E., White, D.C., 1994. Aerobic mineralization of trichloroethylene, vinyl chloride, and aromatic compounds by *Rhodococcus* species. *Appl. Environ. Microbiol.* 60 (2), 542–548.
- Masuda, H., McClay, K., Steffan, R.J., Zylstra, G.J., 2012. Characterization of three propane-inducible oxygenases in *Mycobacterium* sp. strain ENV421. *Lett. Appl. Microbiol.* 55 (3), 175–181.
- McCarty, P.L., 2010. In: F. H., Ward, C.H. (Eds.), *In Situ Remediation of Chlorinated Solvents*. Stroom.
- Menzel, P., Ng, K.L., Krogh, A., 2016. Fast and sensitive taxonomic classification for metagenomics with Kaiju. *Nat. Commun.* 7, 11257.
- Nelson, M.J., Montgomery, S.O., Pritchard, P.H., 1988. Trichloroethylene metabolism by microorganisms that degrade aromatic compounds. *Appl. Environ. Microbiol.* 54 (2), 604–606.
- Parks, D.H., Imelfort, M., Skennerton, C.T., Hugenholtz, P., Tyson, G.W., 2015. CheckM: assessing the quality of microbial genomes recovered from isolates, single cells, and metagenomes. *Genome Res.* 25 (7), 1043–1055.
- Parsons, F., Lage, B.G., 1985. Chlorinated organics in simulated groundwater environments. *Journal of American Water Works Association* 77, 52–59.

- Pfiffner, S.M., Palumbo, A.V., Phelps, T.J., Hazen, T.C., 1997. Effects of nutrient dosing on subsurface methanotrophic populations and trichloroethylene degradation. *J. Ind. Microbiol. Biotechnol.* 18 (2–3), 204–212.
- R Core Team, 2018. R: a Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.
- Ramalingam, V., Cupples, A.M., 2020. Anaerobic 1,4-dioxane biodegradation and microbial community analysis in microcosms inoculated with soils or sediments and different electron acceptors. *Appl. Microbiol. Biotechnol.* 104 (9), 4155–4170.
- Rolston, H., Hyman, M., Semprini, L., 2022. Single-well push-pull tests evaluating isobutane as a primary substrate for promoting in situ cometabolic biotransformation reactions. *Biodegradation* 33, 349–371.
- RStudio Team, 2020. Rstudio. Integrated Development for R. RStudio, PBC, Boston, MA. URL: <http://www.rstudio.com/>.
- Saeki, H., Akira, M., Furuhashi, K., Averhoff, B., Gottschalk, G., 1999. Degradation of trichloroethene by a linear-plasmid-encoded alkene monooxygenase in *Rhodococcus corallinus* (*Nocardia corallina*) B-276. *Microbiology (Read.)* 145 (Pt 7), 1721–1730.
- Seemann, T., 2014. Prokka: rapid prokaryotic genome annotation. *Bioinformatics* 30 (14), 2068–2069.
- Semprini, L., Dolan, M.E., Mathias, M.A., Hopkins, G.H., McCarty, P.L., 2007. Laboratory, Field, and Modeling Studies of Bioaugmentation of butane-utilizing Microorganisms for the in Situ Cometabolic Treatment of 1,1-dichloroethene, 1,1-dichloroethane, and 1,1,1-trichloroethane. 30, 1528–1546.
- Sharp, J.O., Sales, C.M., Alvarez-Cohen, L., 2010. Functional characterization of propane-enhanced N-nitrosodimethylamine degradation by two actinomycetales. *Biotechnol. Bioeng.* 107 (6), 924–932.
- Sharp, J.O., Sales, C.M., LeBlanc, J.C., Liu, J., Wood, T.K., Eltis, L.D., Mohn, W.W., Alvarez-Cohen, L., 2007. An inducible propane monooxygenase is responsible for N-nitrosodimethylamine degradation by *Rhodococcus* sp strain RHA1. *Appl. Environ. Microbiol.* 73, 6930–6938.
- Shen, W., Chen, H., Pan, S., 2008. Anaerobic biodegradation of 1,4-dioxane by sludge enriched with iron-reducing microorganisms. *Bioresour. Technol.* 99 (7), 2483–2487.
- Sieber, C.M.K., Probst, A.J., Sharrar, A., Thomas, B.C., Hess, M., Tringe, S.G., Banfield, J. F., 2018. Recovery of genomes from metagenomes via a dereplication, aggregation and scoring strategy. *Nat. Microbiol.* 3 (7), 836–843.
- Skinner, J., Delgado, A.G., Hyman, M., Chu, M.J., 2024. Implementation of in situ aerobic cometabolism for groundwater treatment: state of the knowledge and important factors for field operation. *Sci. Total Environ.* 925, 171667.
- Steffan, R.J., McClay, K., Vainberg, S., Condee, C.W., Zhang, D., 1997. Biodegradation of the gasoline oxygenates methyl tert-butyl ether, ethyl tert-butyl ether, and tert-amyl methyl ether by propane-oxidizing bacteria. *Appl. Environ. Microbiol.* 63 (11), 4216–4222.
- Steffan, R.J., Vainberg, S., 2013. In: Stroo, H.F., Leeson, A., Ward, C.H. (Eds.), *Bioaugmentation for Groundwater Remediation*. Springer, New York, pp. 89–115.
- Suttinun, O., Muller, R., Luepromchai, E., 2010. Cometabolic degradation of trichloroethene by *Rhodococcus* sp. strain L4 immobilized on plant materials rich in essential oils. *Appl. Environ. Microbiol.* 76 (14), 4684–4690.
- Thompson, J.D., Higgins, D.G., Gibson, T.J., 1994. Clustal W: improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice. *Nucleic Acids Res.* 22, 4673–4680.
- Vogel, T.M., McCarty, P.L., 1985. Biotransformation of tetrachloroethylene to trichloroethylene, dichloroethylene, vinyl chloride, and carbon dioxide under methanogenic conditions. *Appl. Environ. Microbiol.* 49 (5), 1080–1083.
- Vogel, T.M., McCarty, P.L., 1987. Abiotic and biotic transformations of 1,1,1-trichloroethane under methanogenic conditions. *Environ. Sci. Technol.* 21, 1208–1213.
- Wackett, L.P., 1996. Co-metabolism: is the emperor wearing any clothes? *Curr. Opin. Biotechnol.* 7 (3), 321–325.
- Wackett, L.P., Brusseau, G.A., Householder, S.R., Hanson, R.S., 1989. Survey of microbial oxygenases: trichloroethylene degradation by propane-oxidizing bacteria. *Appl. Environ. Microbiol.* 55 (11), 2960–2964.
- Wang, B., Chu, K.H., 2017. Cometabolic biodegradation of 1,2,3-trichloropropane by propane-oxidizing bacteria. *Chemosphere* 168, 1494–1497.
- Wickham, H., 2016. *ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag, New York, 2016. <https://ggplot2.tidyverse.org>.
- Wickham, H., Bryan, J., 2023. Readxl: read Excel files. R package, version 1.4.2. <https://CRAN.R-project.org/package=readxl>.
- Wilkins, D., 2023. Gggenes: draw Gene Arrow Maps in 'ggplot2'. R package version 0.5.1. <https://CRAN.R-project.org/package=gggenes>.
- Wilson, J.T., Wilson, B.H., 1985. Biotransformation of trichloroethylene in soil. *Appl. Environ. Microbiol.* 49 (1), 242–243.
- Wu, Y.W., Simmons, B.A., Singer, S.W., 2016. MaxBin 2.0: an automated binning algorithm to recover genomes from multiple metagenomic datasets. *Bioinformatics* 32 (4), 605–607.
- Yang, S.N.N., Haritos, V., Kertesz, M.A., Coleman, N.V., 2024. A novel soluble di-iron monooxygenase from the soil bacterium *Solimonas soli*. *Environ. Microbiol.* 26 (2), e16567.
- Zalesak, M., Ruzicka, J., Vicha, R., Dvorackova, M., 2017. Cometabolic degradation of dichloroethenes by *Comamonas testosteroni* RF2. *Chemosphere* 186, 919–927.